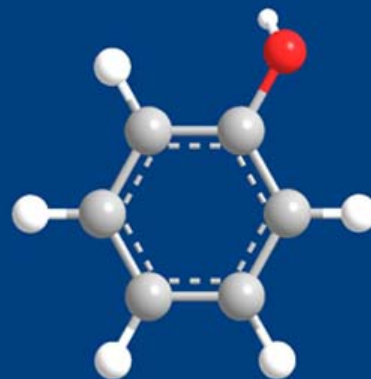
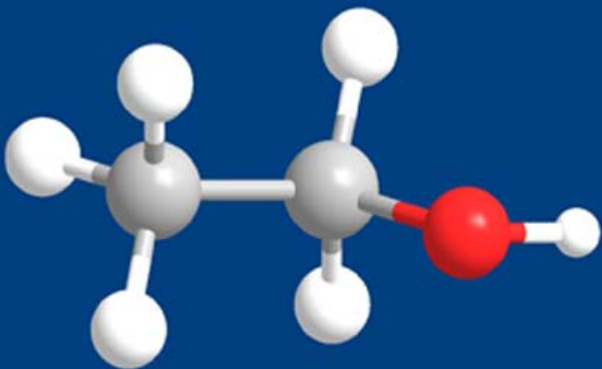


第九章 醇 酚 醚

Alcohols, Phenols and Ethers

厦门大学有机化学B

CONTENT

1

醇的分类、结构、命名和物理性质

2

醇的制备和来源

3

醇的化学性质

4

酚的结构、命名、来源和物理性质(自学)

5

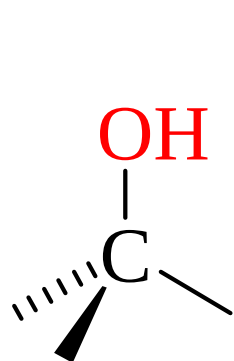
酚的化学性质

6

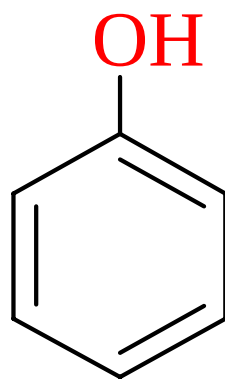
醚

9.1 醇的分类、结构、命名和物理性质

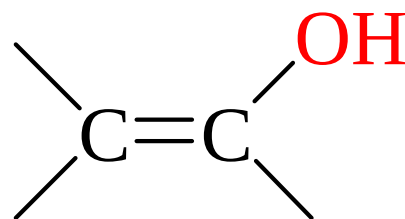
醇是羟基连接在饱和 sp^3 碳原子上的化合物



醇



酚



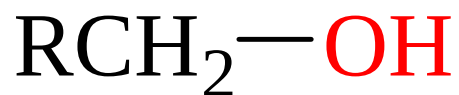
烯醇

- 醇可看成是水分子中的一个氢原子被烃基取代所形成的有机衍生物
- 醇的官能团是羟基(hydroxyl group) —OH

9.1.1 醇的分类

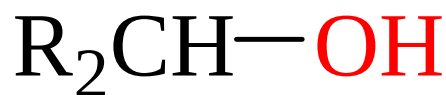
3种分类方法:

1) 根据官能团所连烃基类型



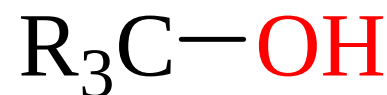
伯醇

一级醇(1°)



仲醇

二级醇(2°)



叔醇

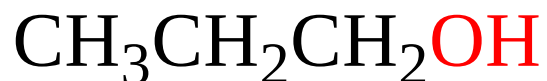
三级醇(3°)

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9.1.1 醇的分类

2) 根据烃基结构

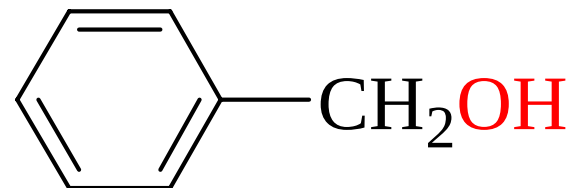
饱和醇



不饱和醇

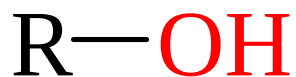


芳香醇

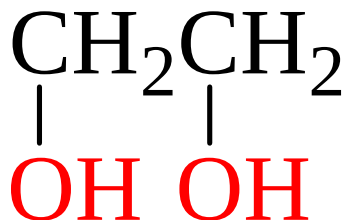


3) 根据羟基数目

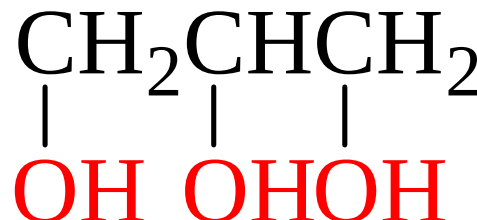
一元醇



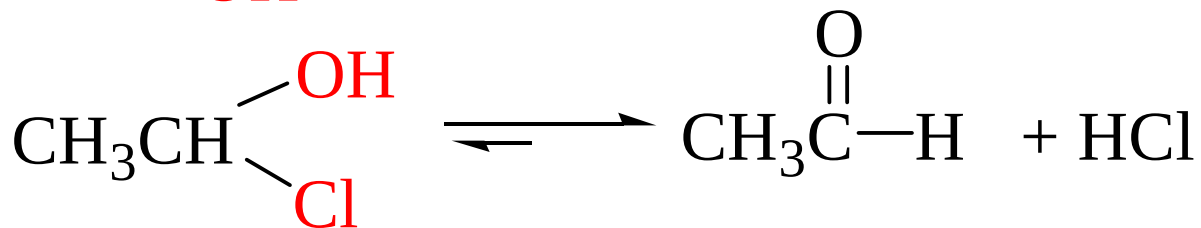
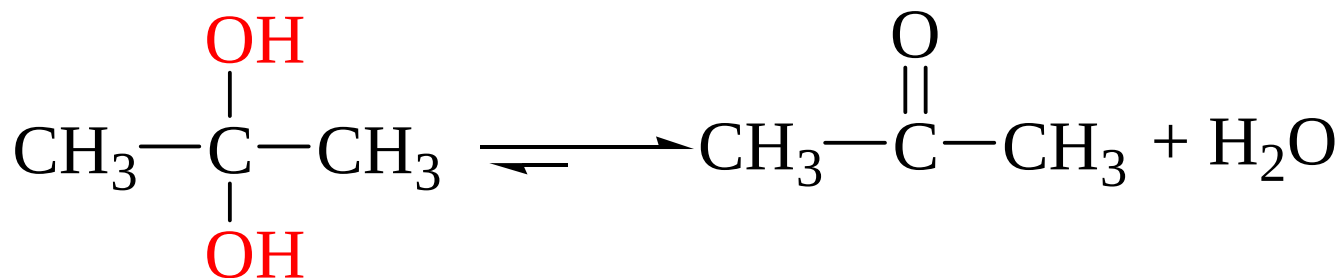
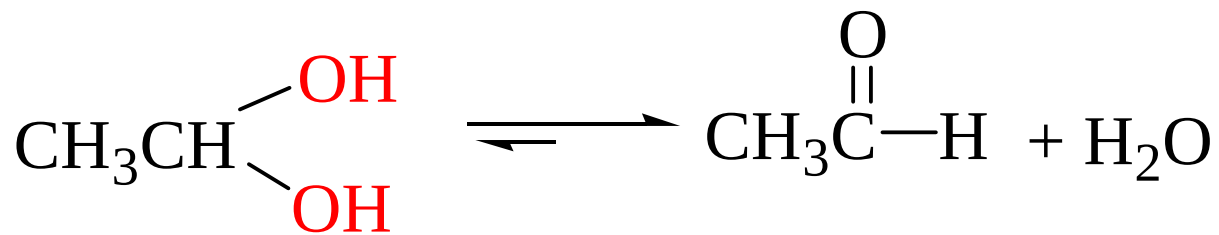
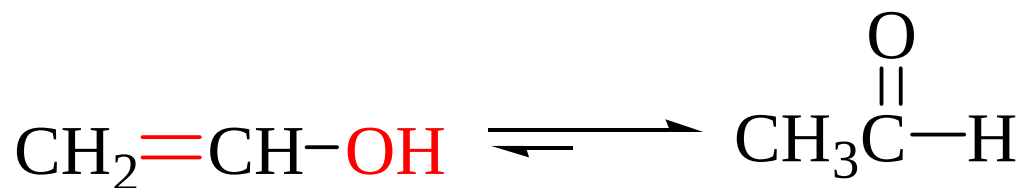
二元醇



多元醇

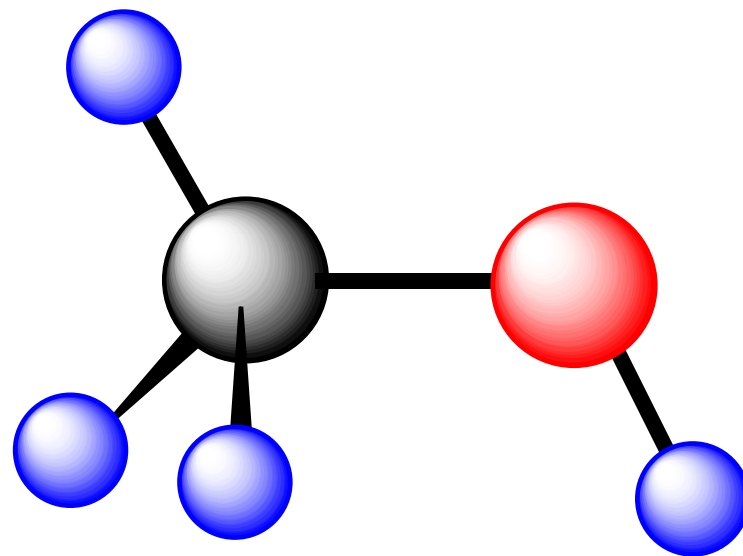
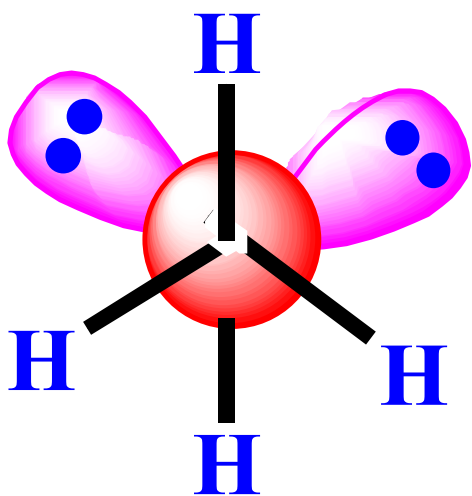


不稳定的醇



9.1.2. 醇的结构

C、O都是 sp^3 杂化



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9.1.3 醇的命名

1) 普通命名

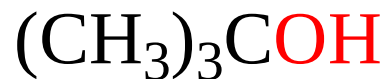
中文：将相应烷烃名称中的烷→醇



正丁醇



异丁醇

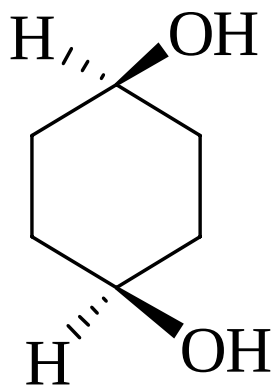


叔丁醇

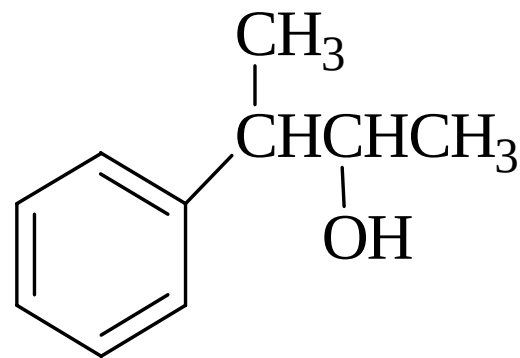
厦门大学有机化学B

9.1.3 醇的命名

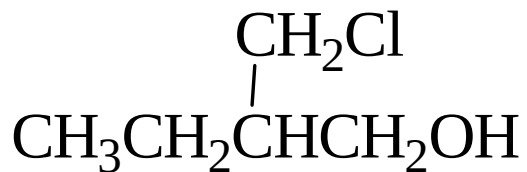
2) 系统命名



顺-1,4-环己二醇



3-苯基-2-丁醇



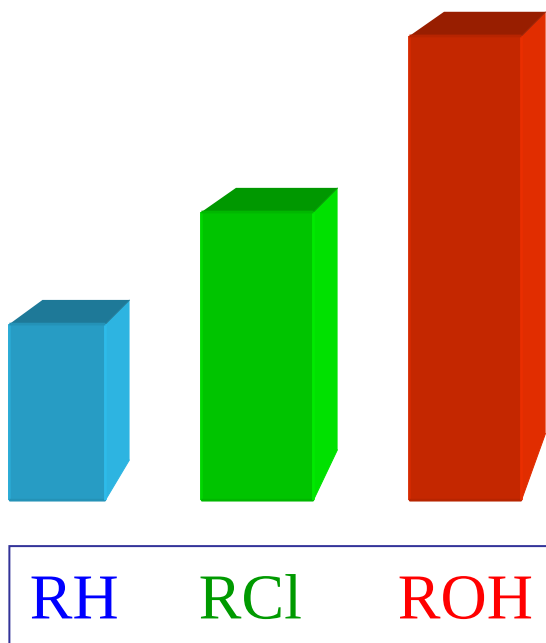
2-(氯甲基)-1-丁醇

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9.1.4 物理性质

- 醇的沸点

比同碳原子数的烷烃、卤代烃高，也比分子量相近的烷烃高

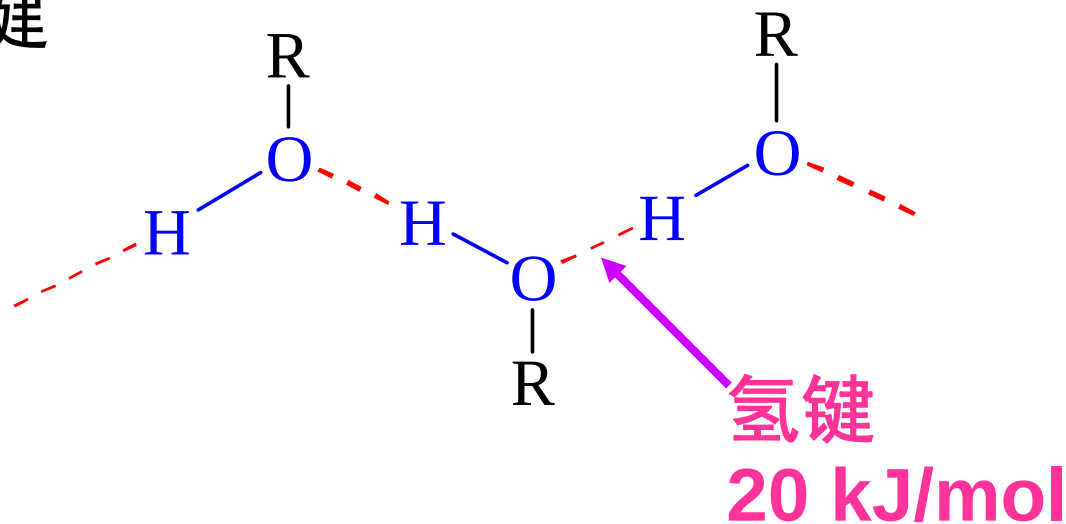


化合物	分子量	沸点(°C)
CH ₃ CH ₂ OH	46	78.5
CH ₃ CH ₃	30	—98
CH ₃ CH ₂ CH ₃	44	—42
CH ₃ CH ₂ Cl	64.5	12

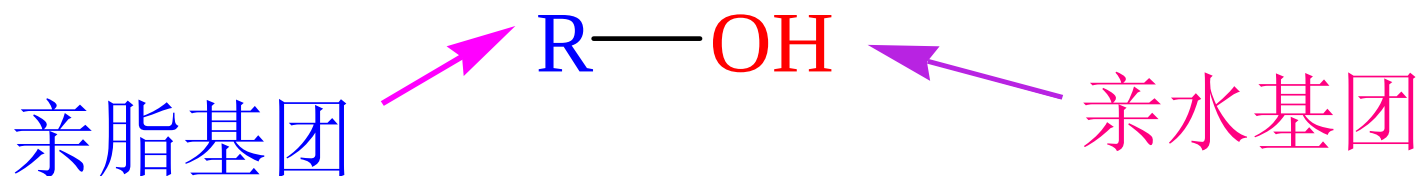
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9.1.4 物理性质

- 醇的分子间氢键



- 溶解度：低级醇溶于水，随分子量增加溶解度降低

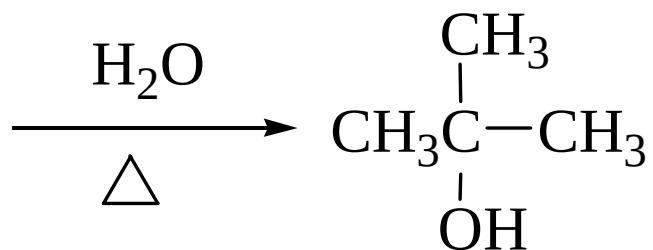
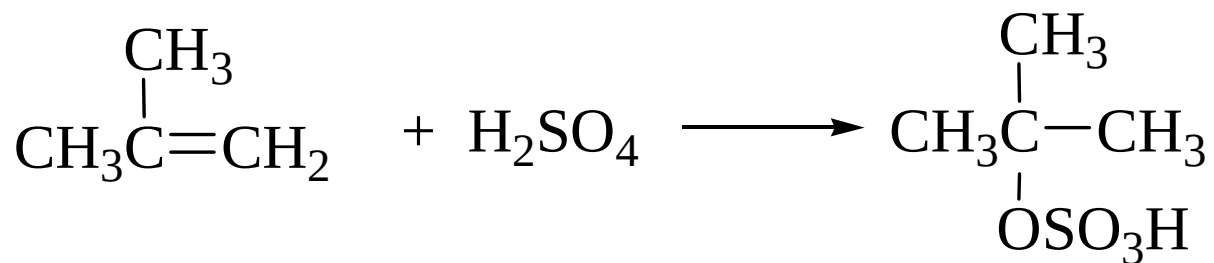
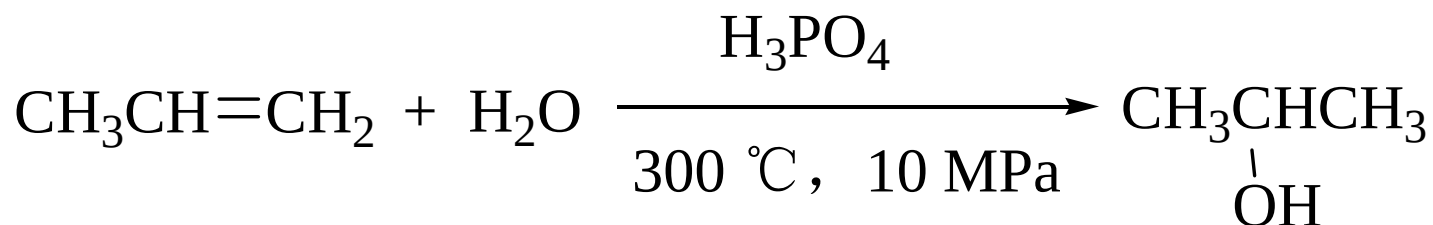


9.2 醇的制备和来源

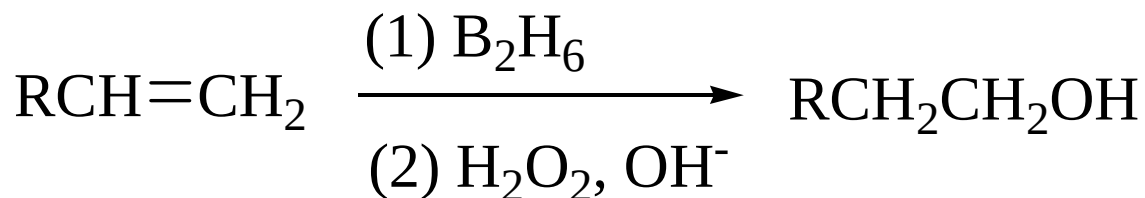
- 由烯烃制备——水合，硼氢化-氧化
- 由羰基化合物制备——Grignard 反应，与炔化物反应，还原
- 由卤代烃水解

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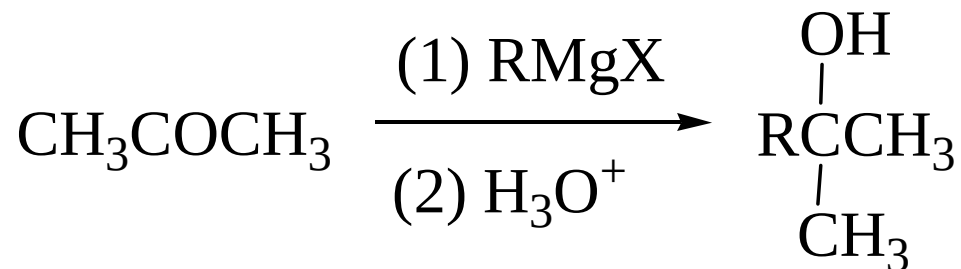
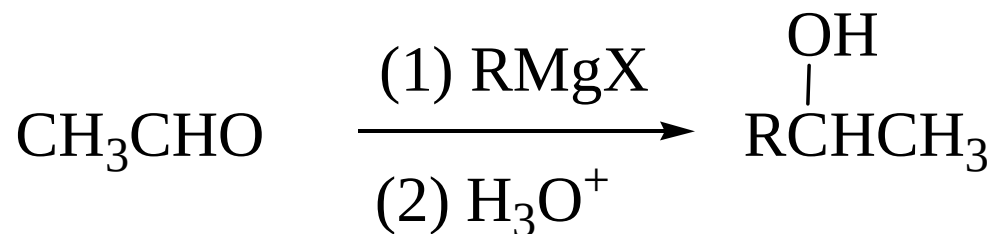
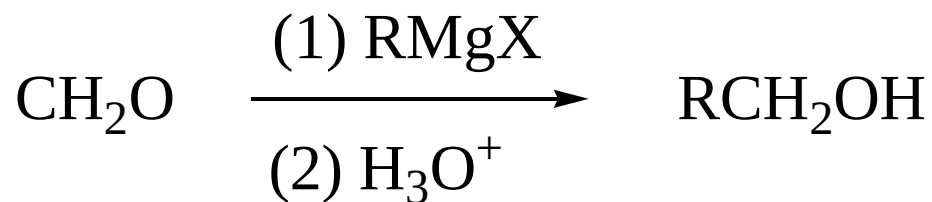
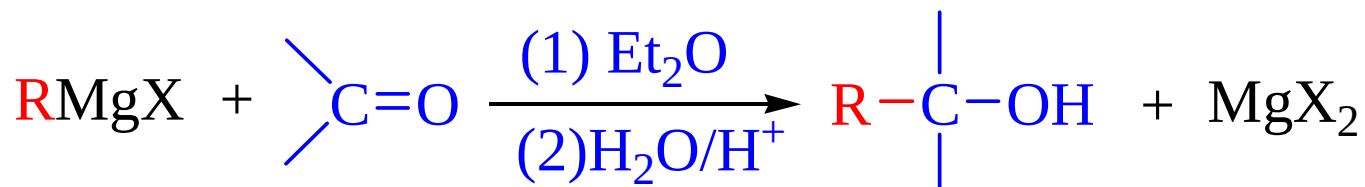
9.2.1 由烯烃制备



可能发生重排

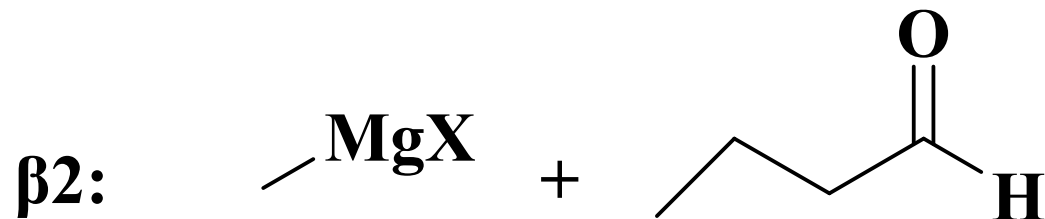
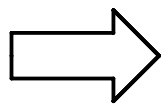
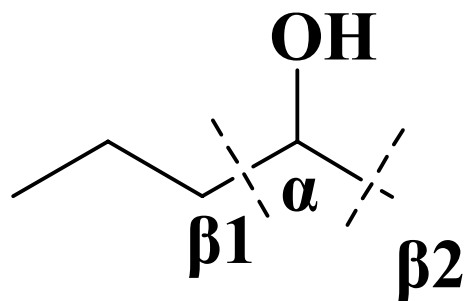
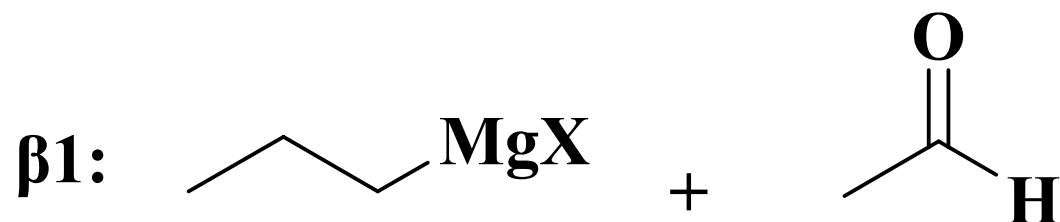


9.2.2 由羰基化合物制备

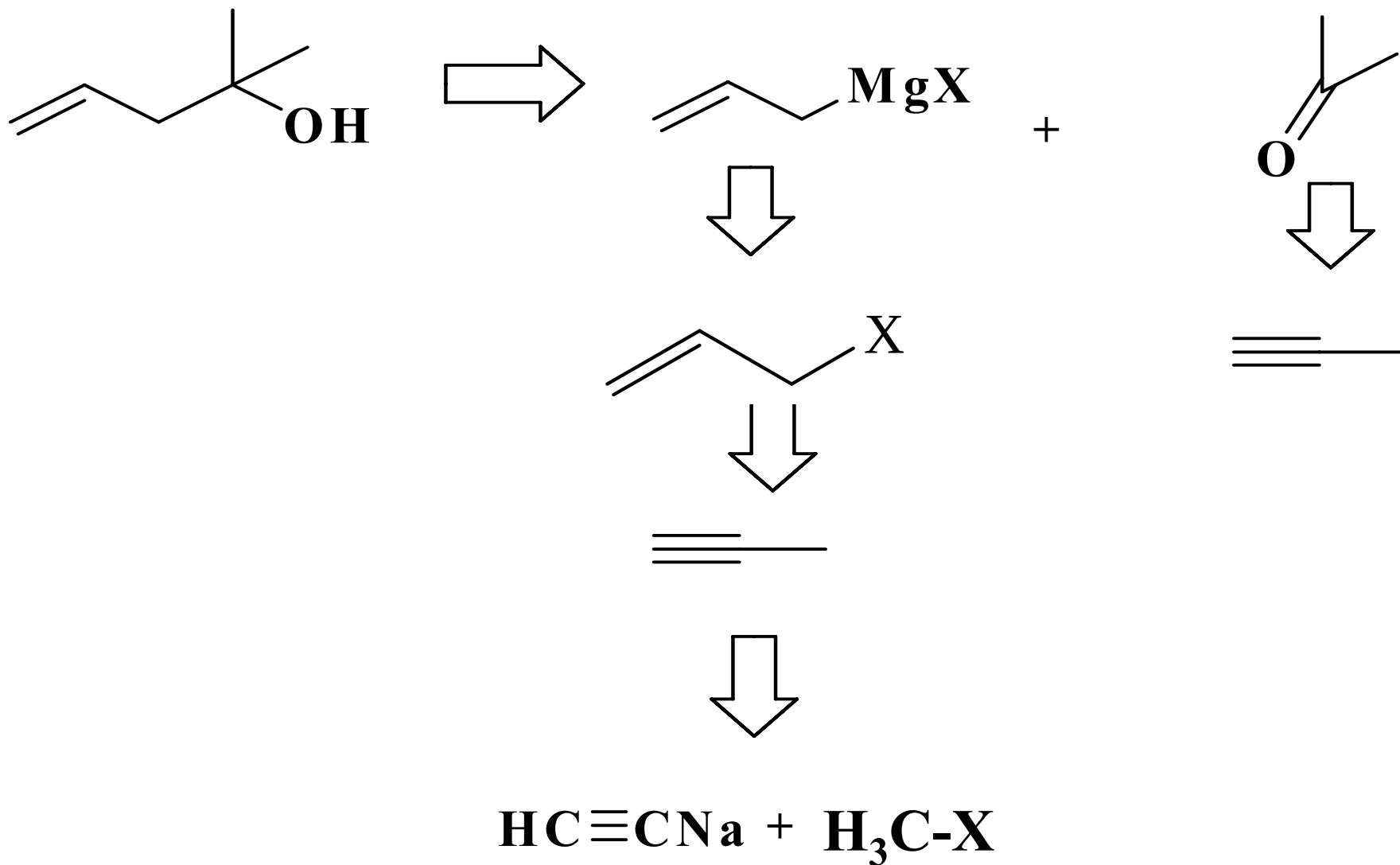


Example

逆向合成分析



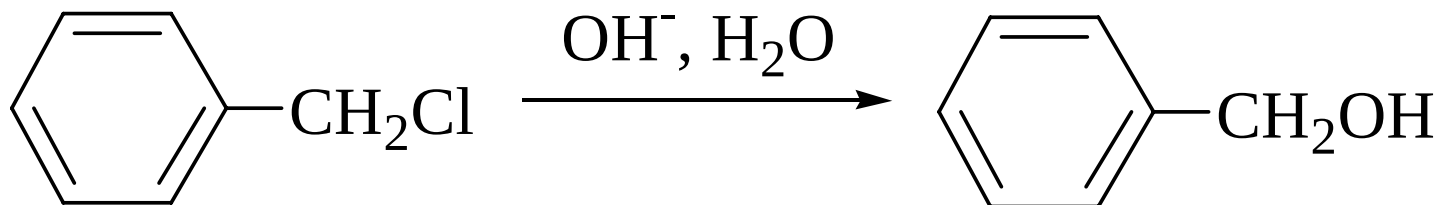
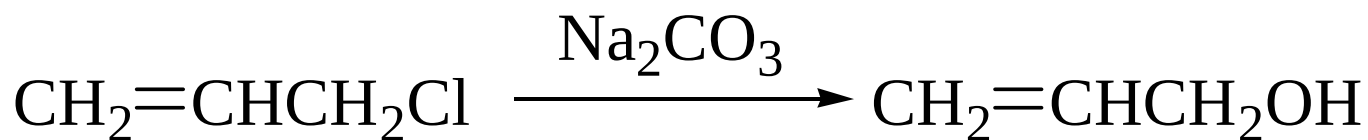
Example



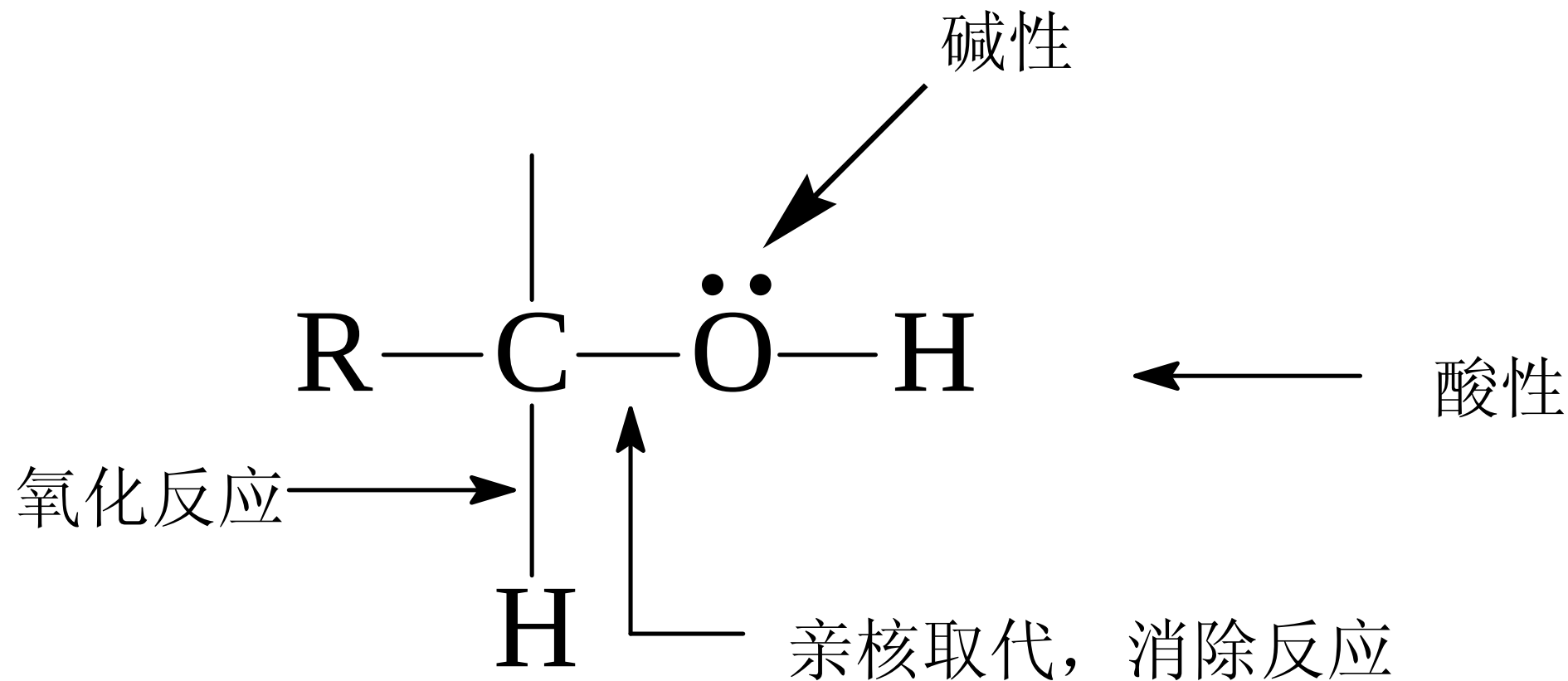
9.2.3 由卤代烃水解

- 副反应多
- 一般情况下醇比卤代烃易得

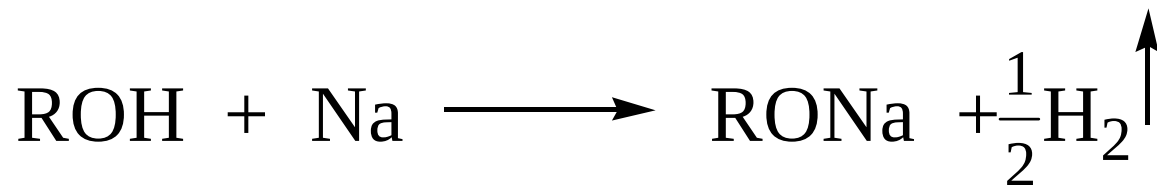
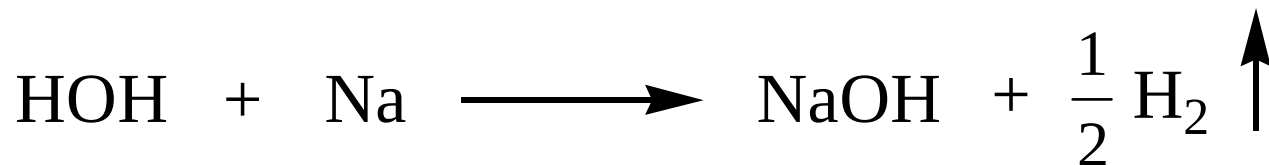
由卤代烃水解主要应用到如下类型的化合物中：



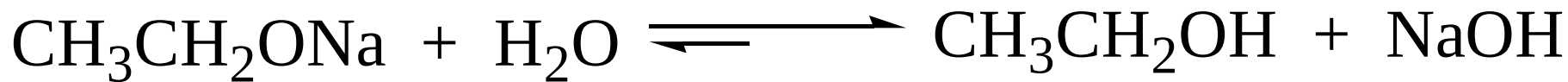
9.3 醇的化学性质



9.3.1 醇的酸性



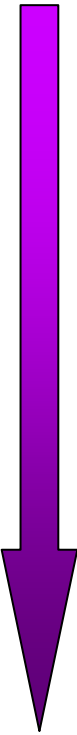
反应速度 $\text{CH}_3\text{OH} > \text{CH}_3\text{CH}_2\text{OH} > \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} > (\text{CH}_3)_2\text{CHOH} > (\text{CH}_3)_3\text{COH}$



醇是一种弱酸

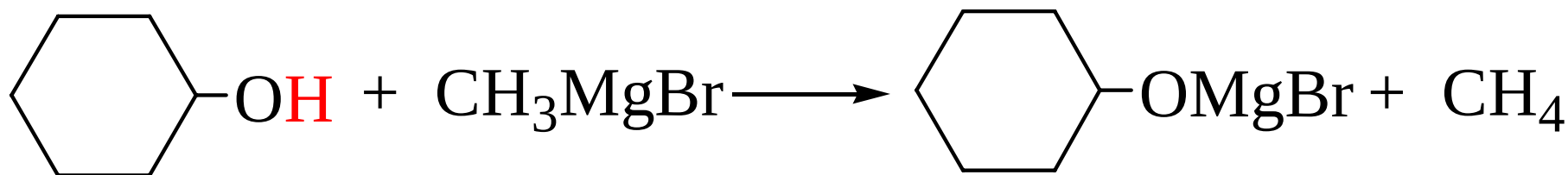
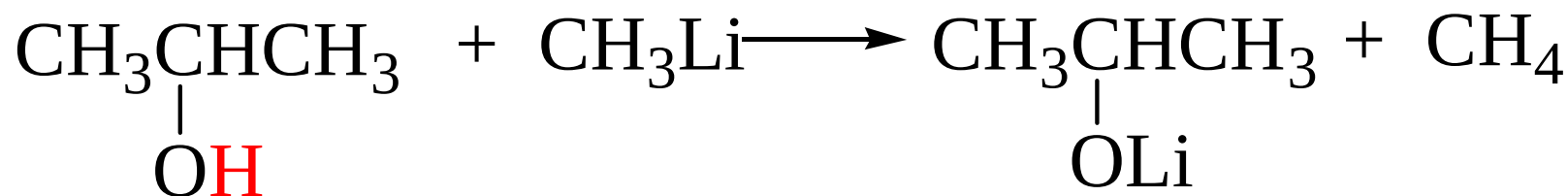
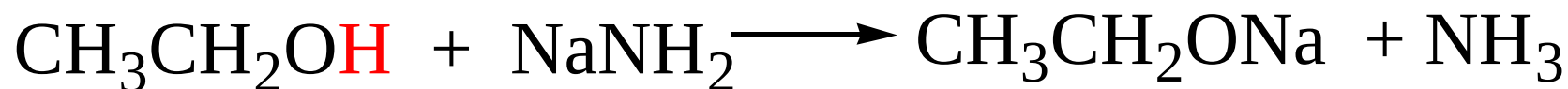
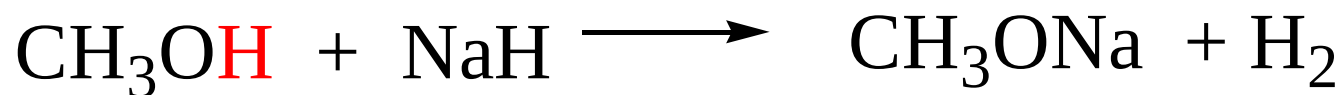
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酸性大小

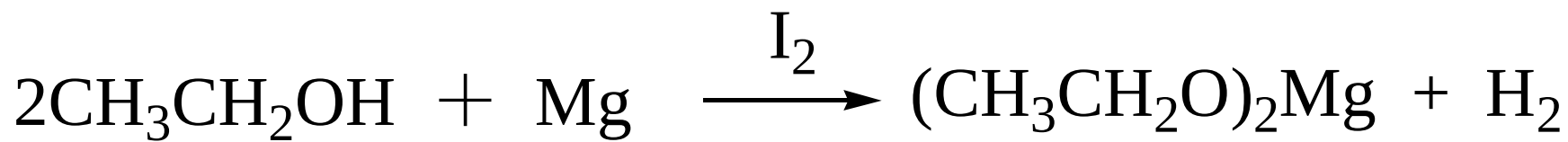
化合物	pK _a	
(CH ₃) ₃ COH	18.00	弱酸
CH ₃ CH ₂ OH	16.00	
HOH	15.74	
CH ₃ OH	15.54	
CF ₃ CH ₂ OH	12.43	
(CF ₃) ₃ COH	5.4	
HCl	—7.00	强酸

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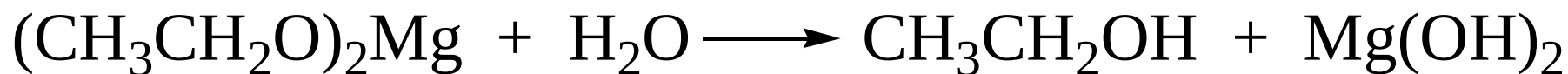
Example



醇与Mg反应



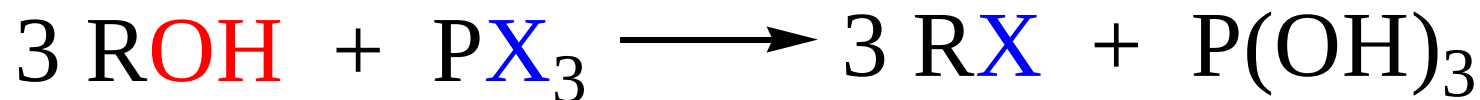
乙醇镁



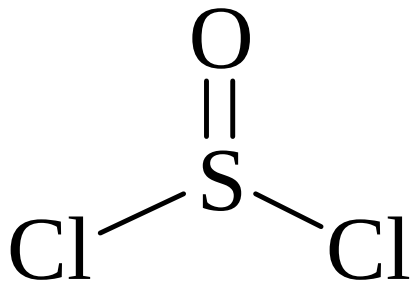
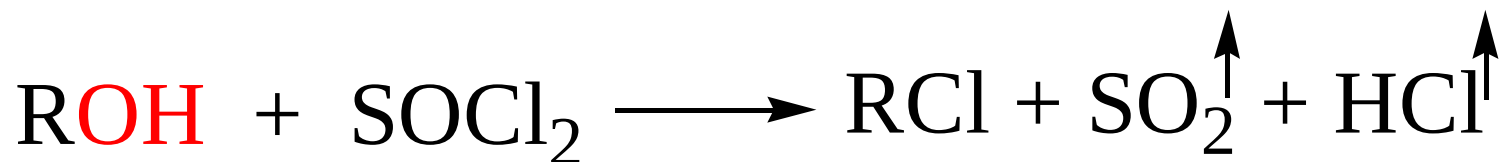
用于制备高度无水乙醇

9.3.2 取代反应

1. 与三卤化磷和氯化亚砷反应



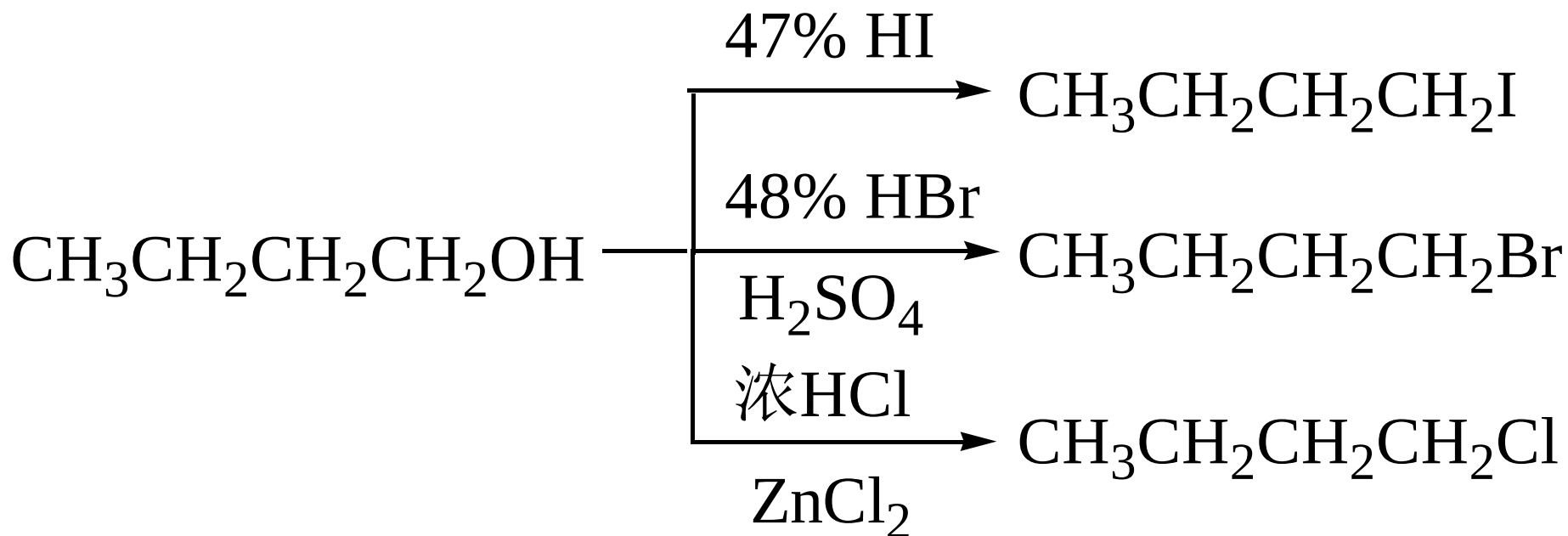
或 PX_5 或 X_2 , P $\text{X} = \text{Br}, \text{I}$



氯化亚砷
亚硫酸氯

9.3.2 取代反应

2. 与氢卤酸反应

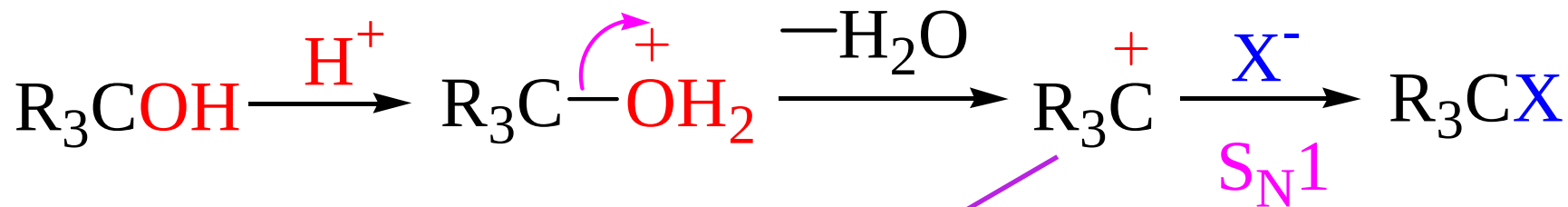
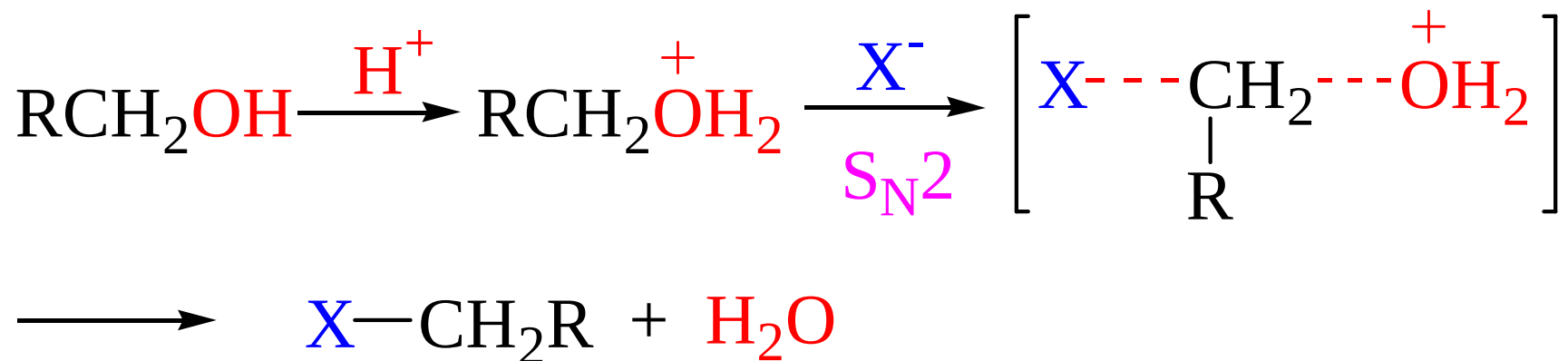


反应活性: $\text{HI} > \text{HBr} > \text{HCl}$

$3^\circ \text{ 醇} > 2^\circ \text{ 醇} > 1^\circ \text{ 醇}$

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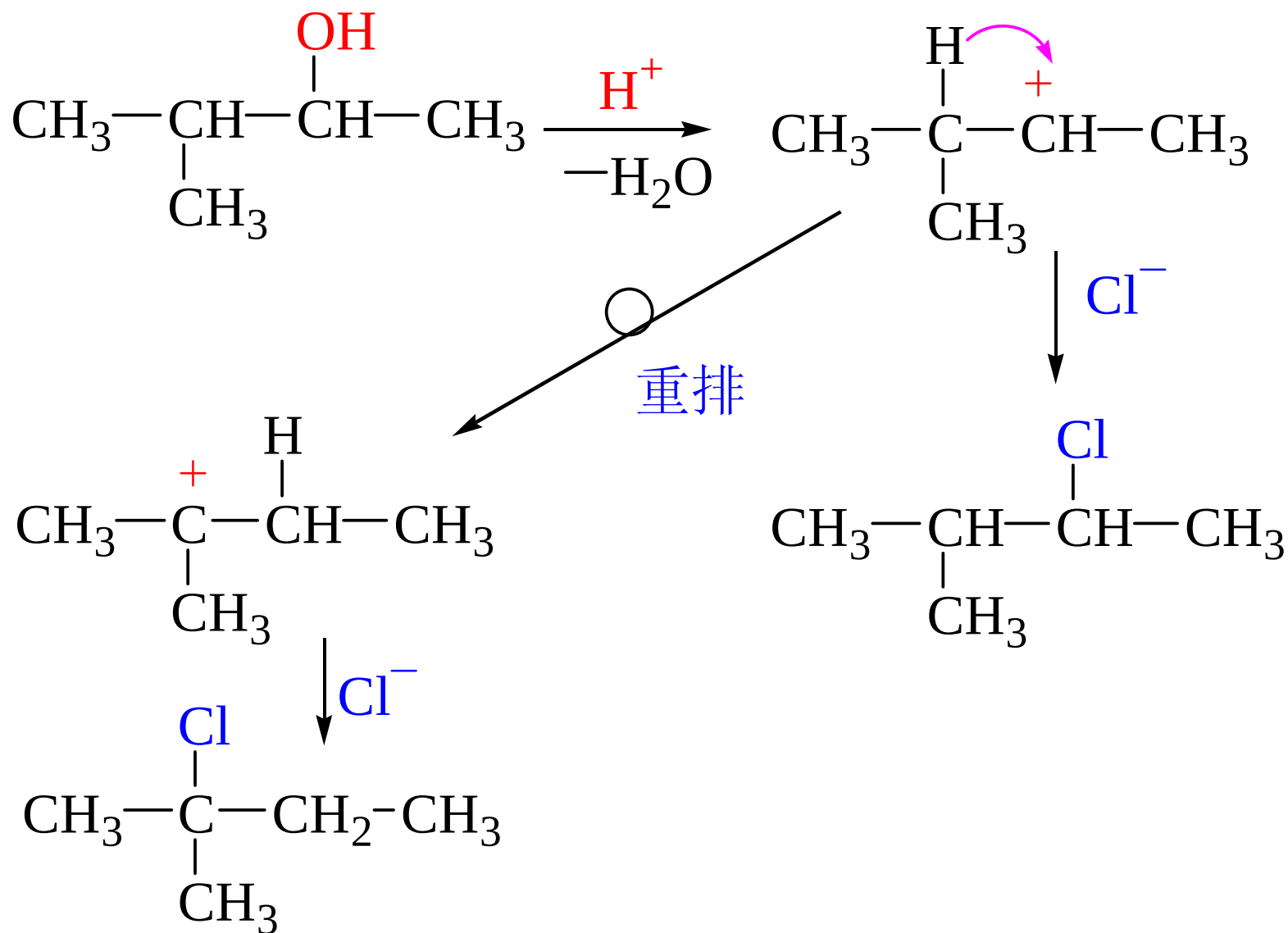
反应机理



重排, 消除.....

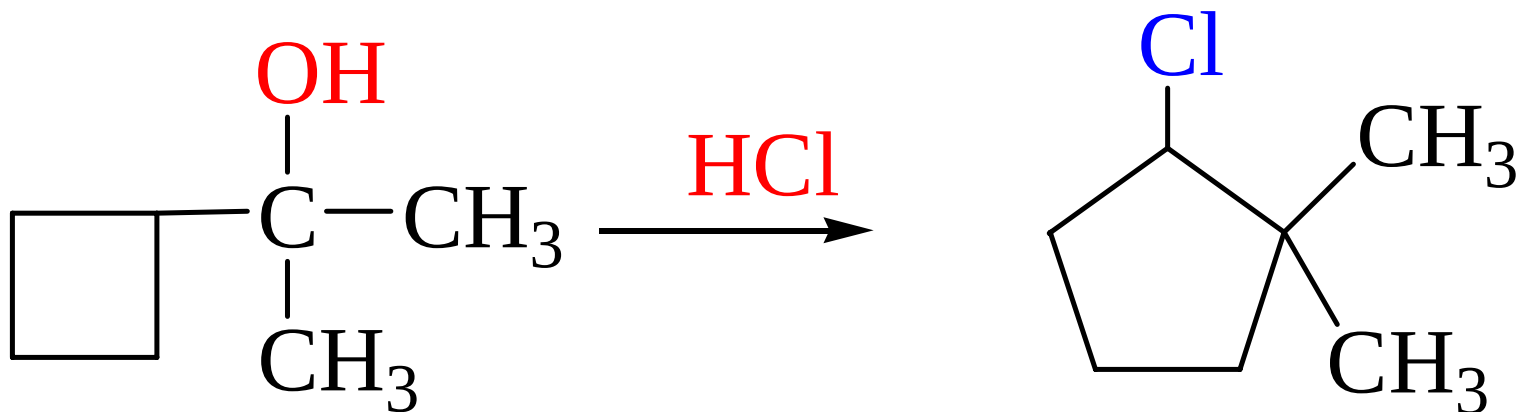
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反应机理

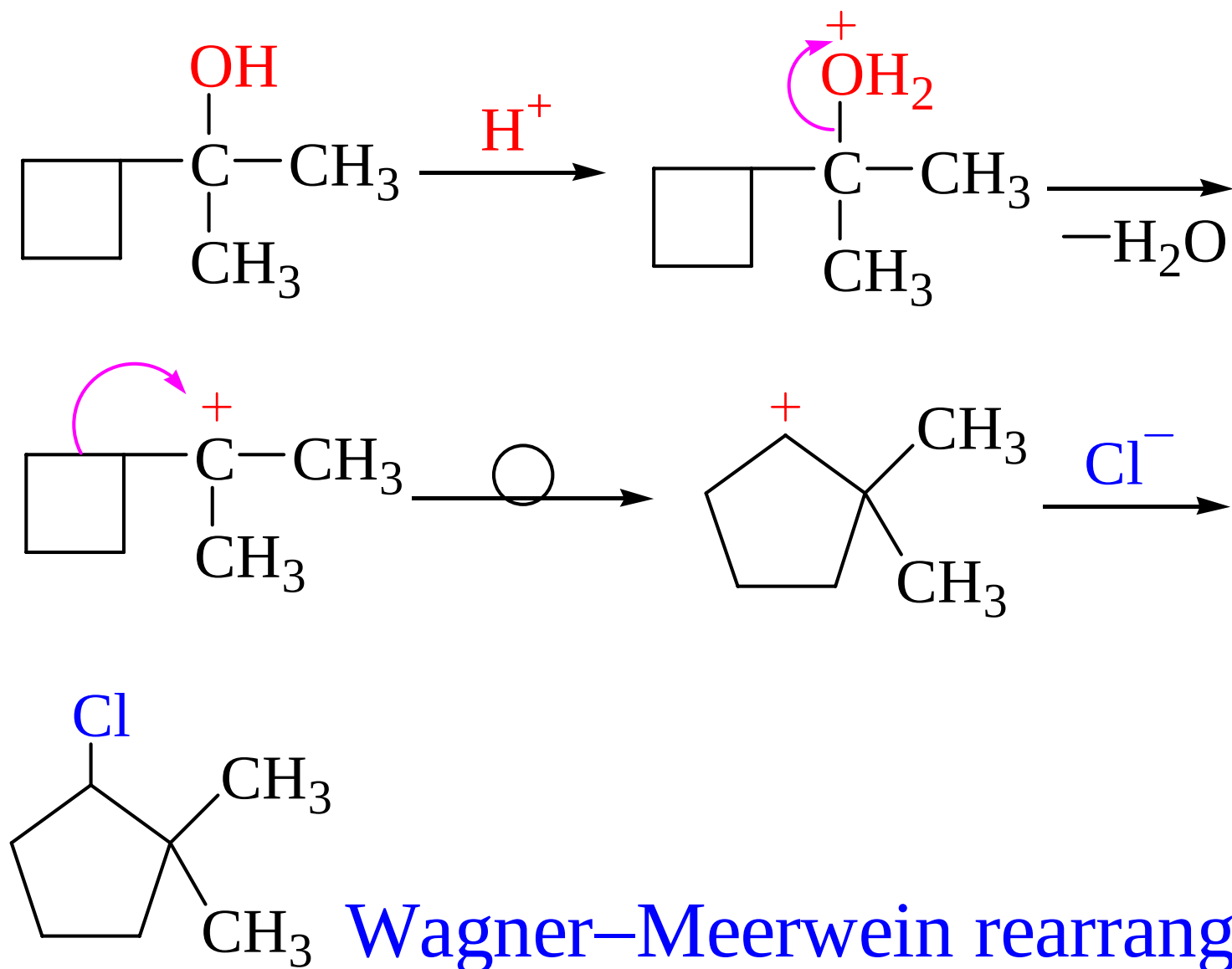


Example

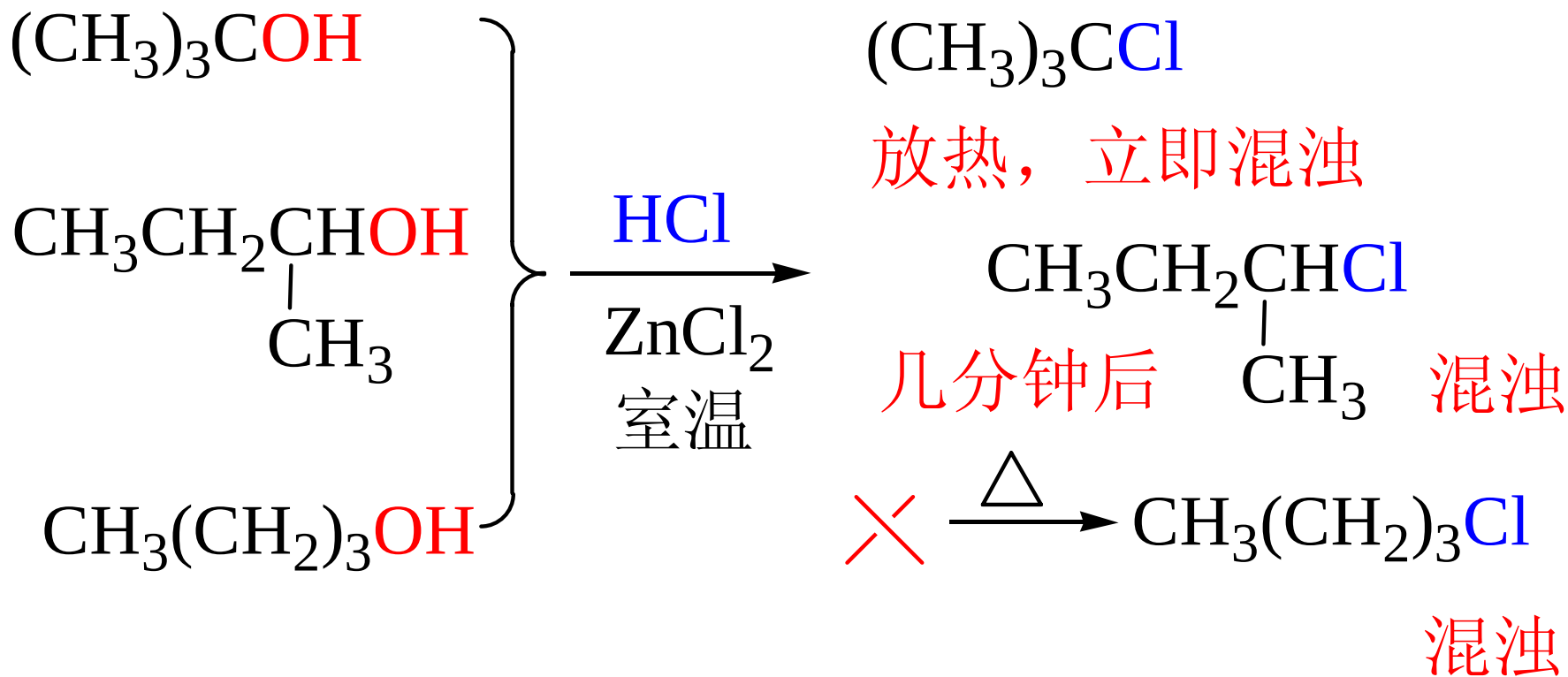
请同学们用机理
解释如下反应



Example



Lucas Reagent (浓HCl/ZnCl₂)

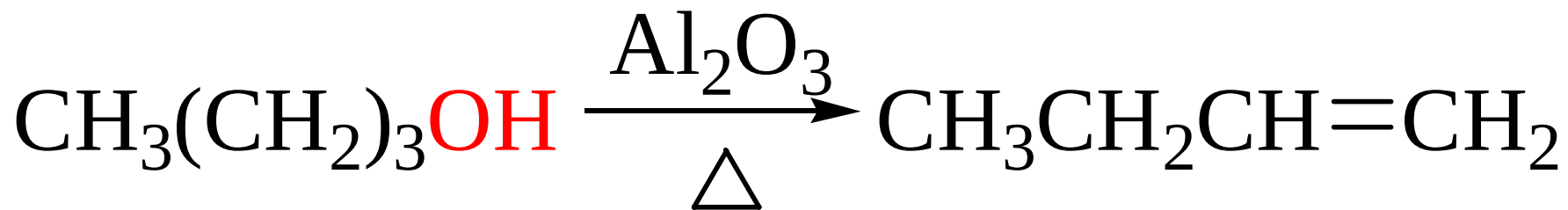


用于伯、仲、叔醇的鉴别
适用于C₄ ~ C₈的醇

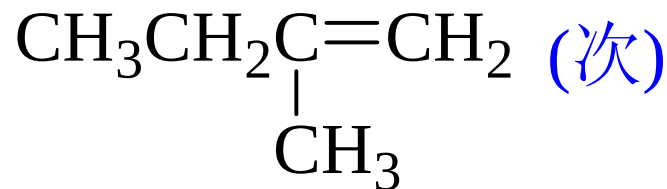
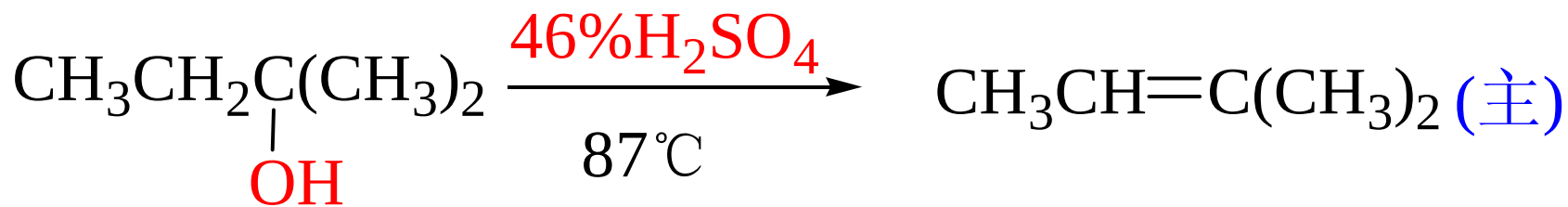
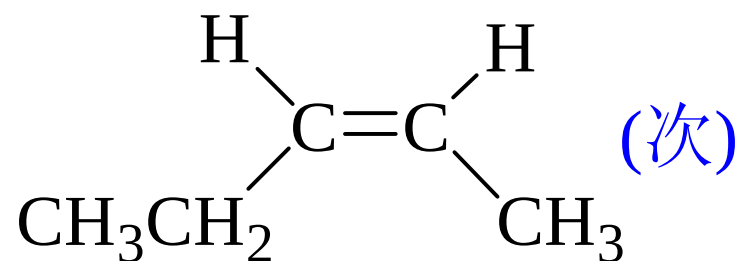
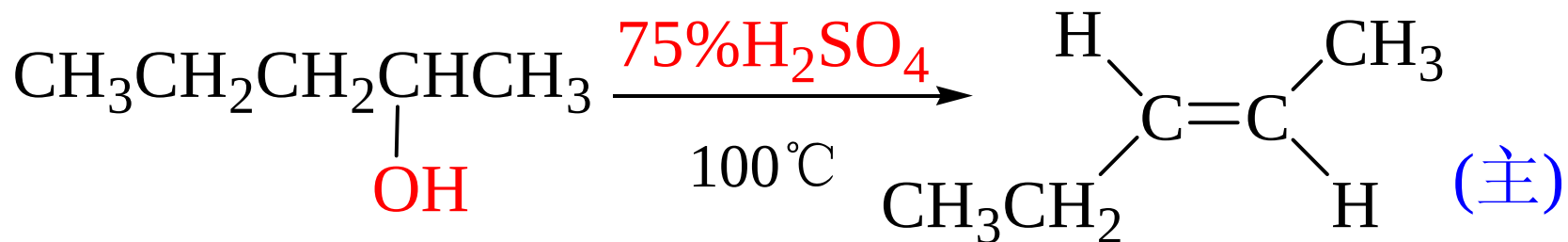
9.3.3 脱水反应

常用催化剂：H₂SO₄, H₃PO₄, Al₂O₃

1. 分子内脱水



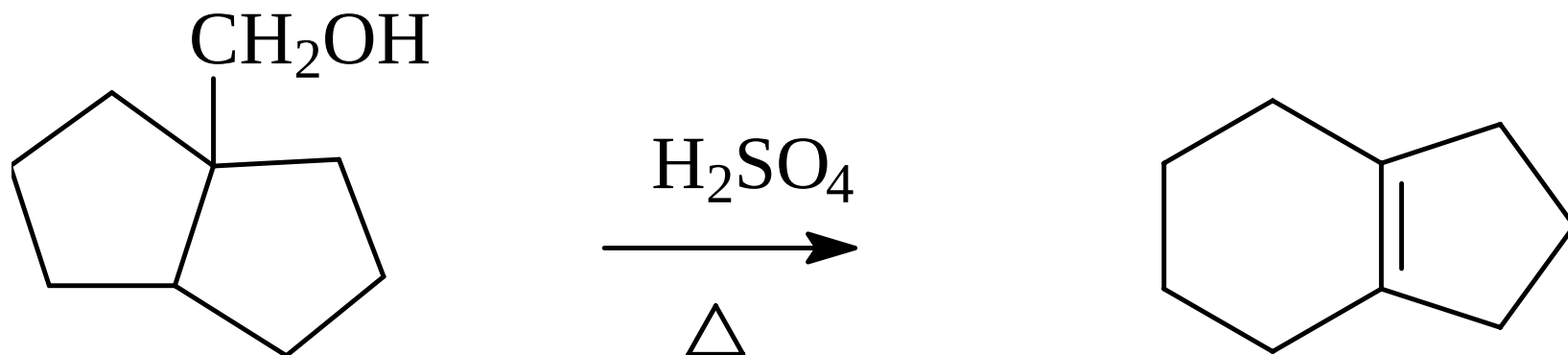
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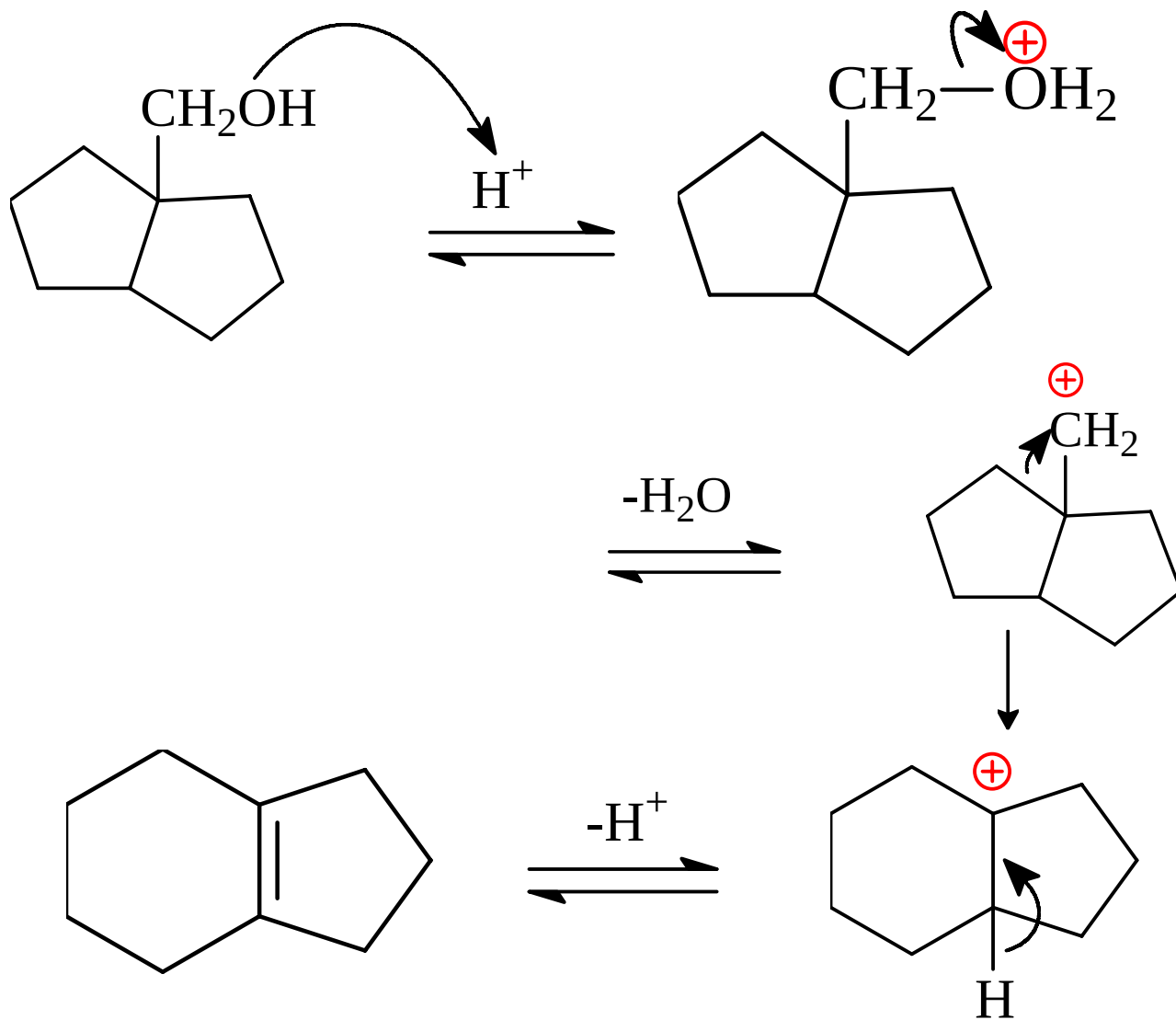
遵从Zaitsev 规律

Example

请同学们用机理
解释如下反应



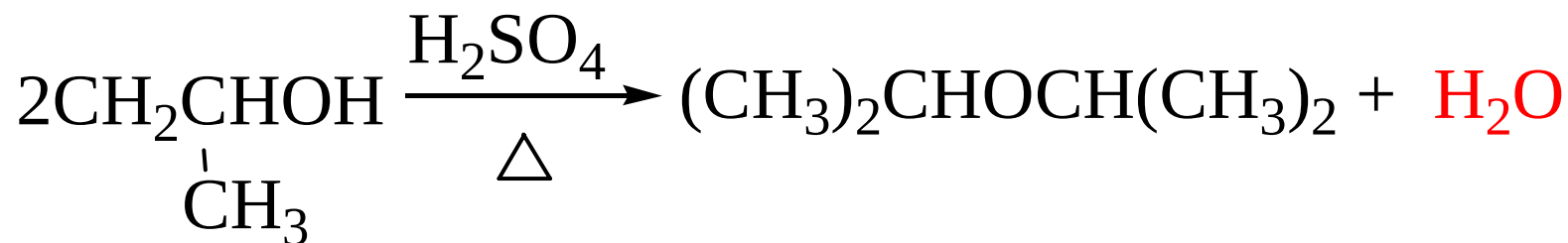
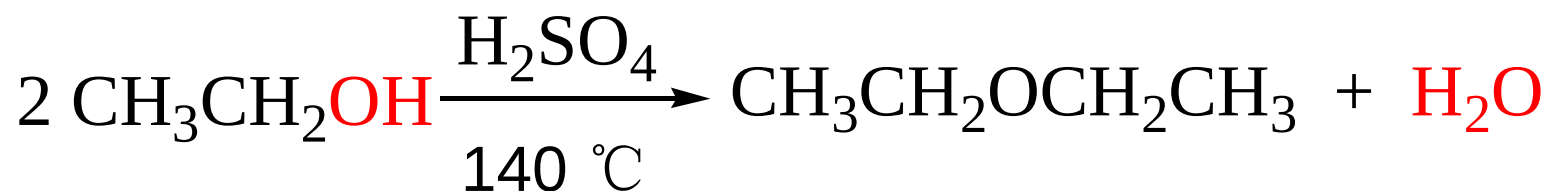
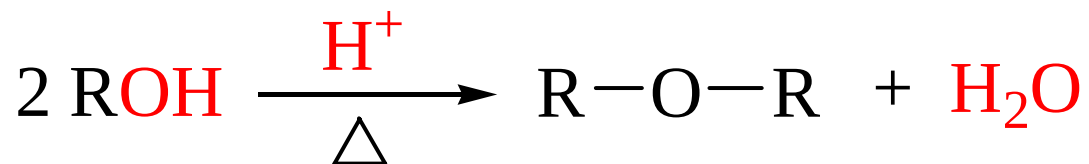
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9.3.3 脱水反应

2. 分子间脱水

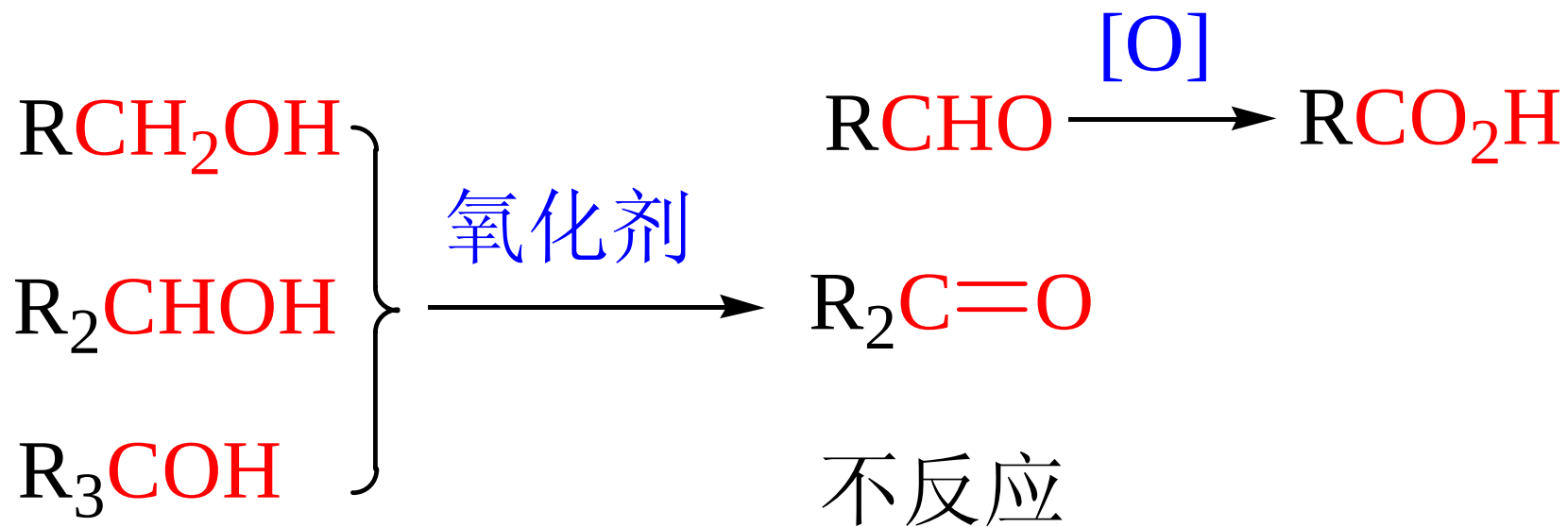


主要副反应：分子内脱水。
不适用于三级醇

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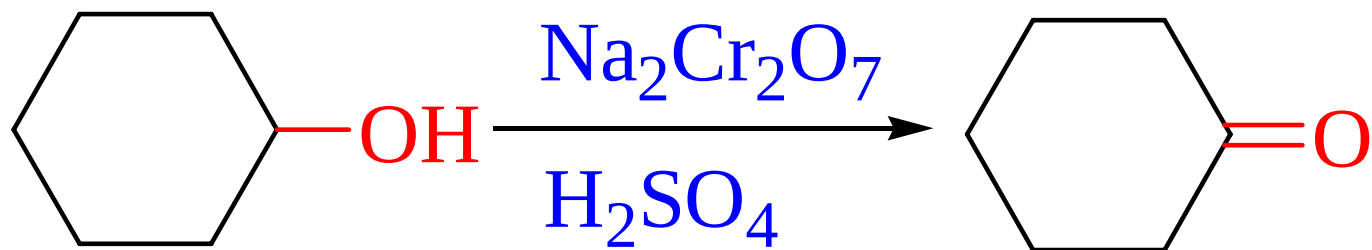
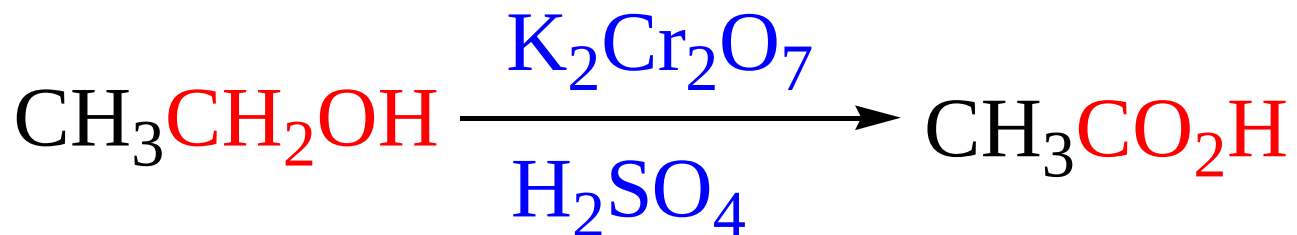
9.3.4 氧化与脱氢

- 氧化剂氧化

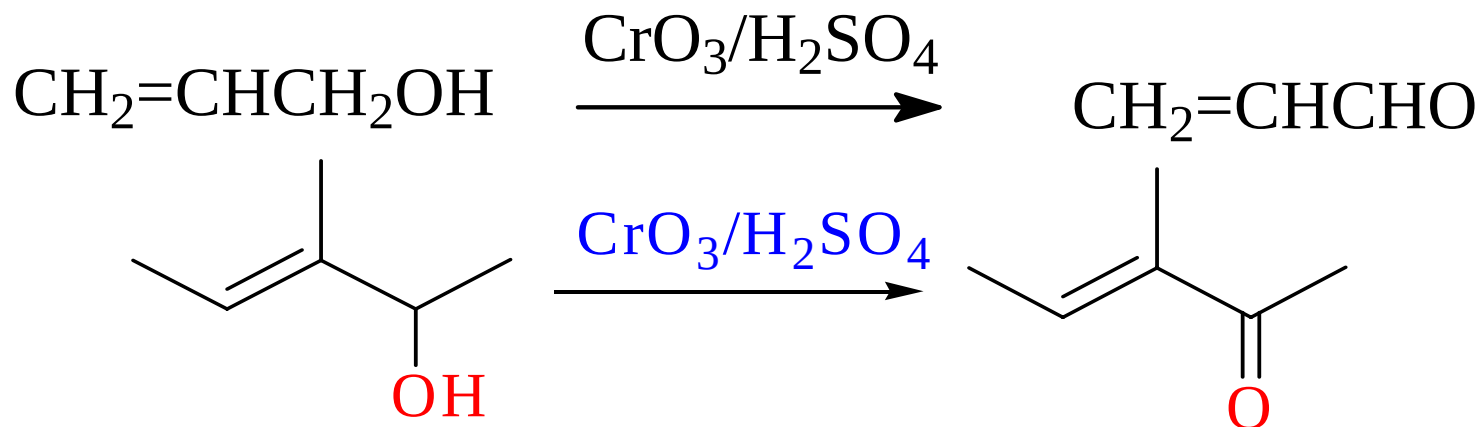


氧化剂: KMnO_4 , H_2CrO_4 , $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$
 $\text{Na}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$, $\text{CrO}_3/\text{H}_2\text{SO}_4$

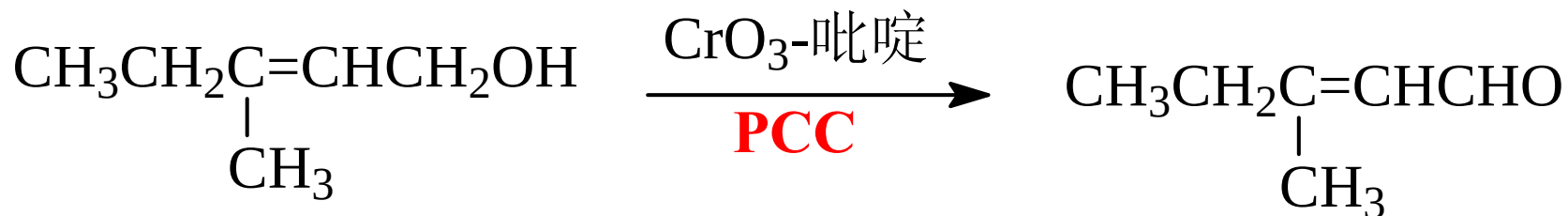
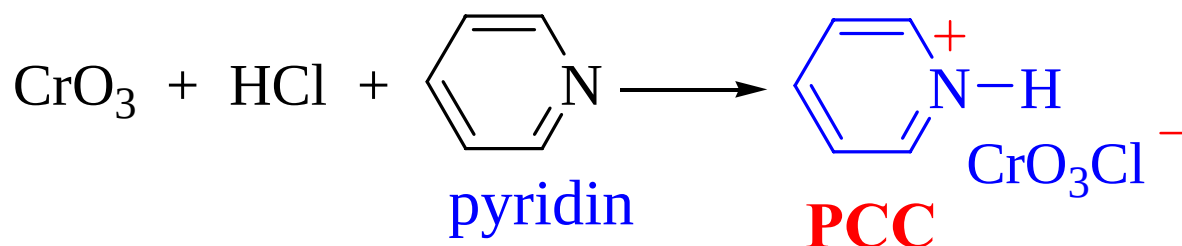
Example



Johns试剂 :CrO₃/H₂SO₄

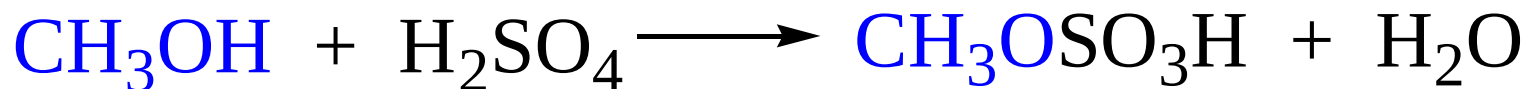
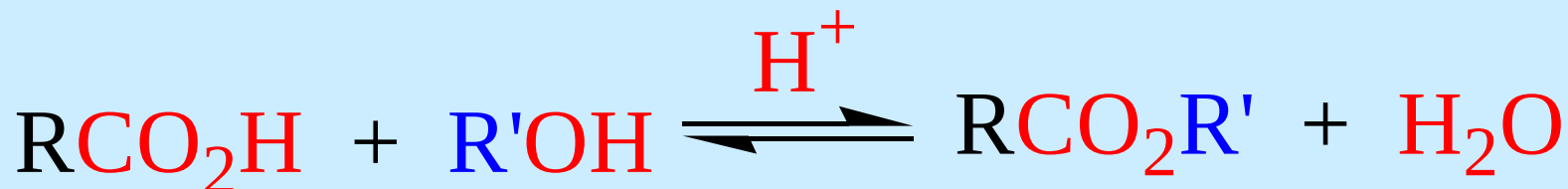


PCC (Pyridinium Chlorochromate)

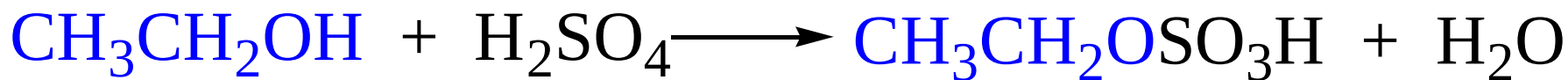


9.3.5 成酯反应

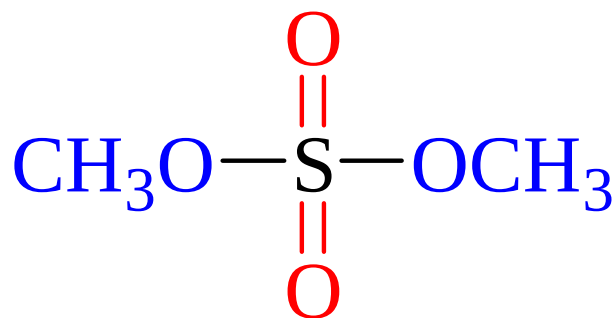
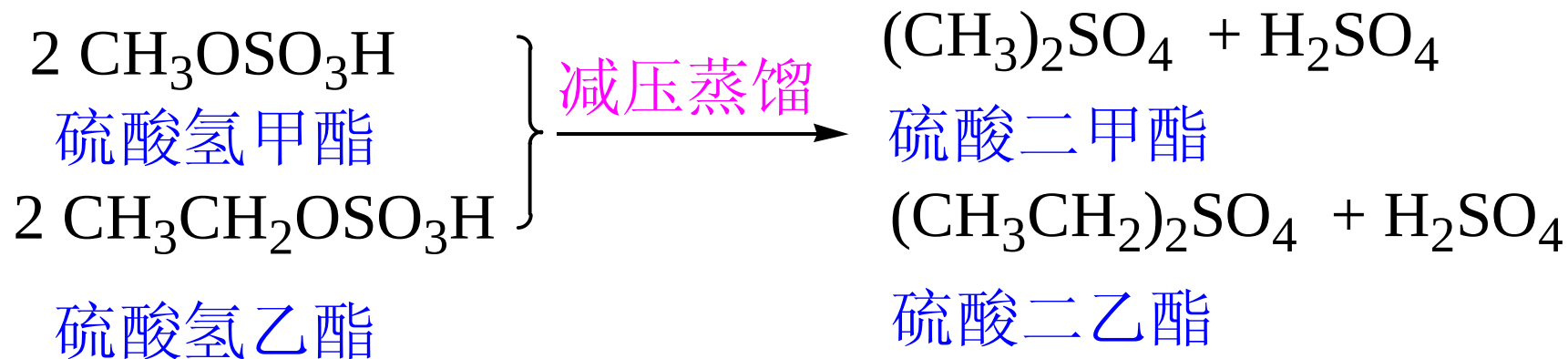
醇与含氧酸(有机酸、无机酸)反应成酯



硫酸氢甲酯



硫酸氢乙酯

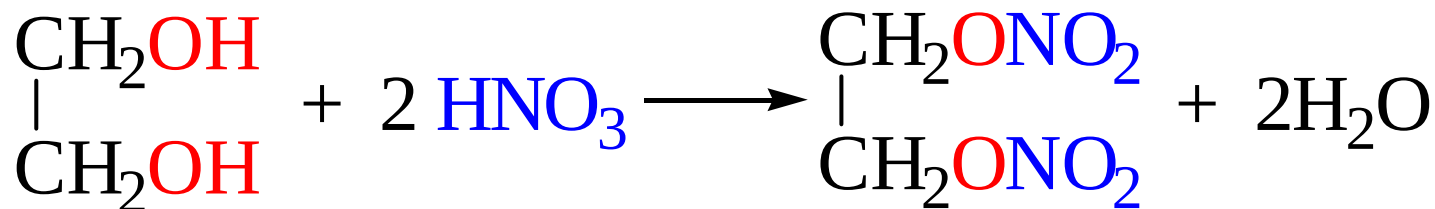


烷基化试剂

剧毒!

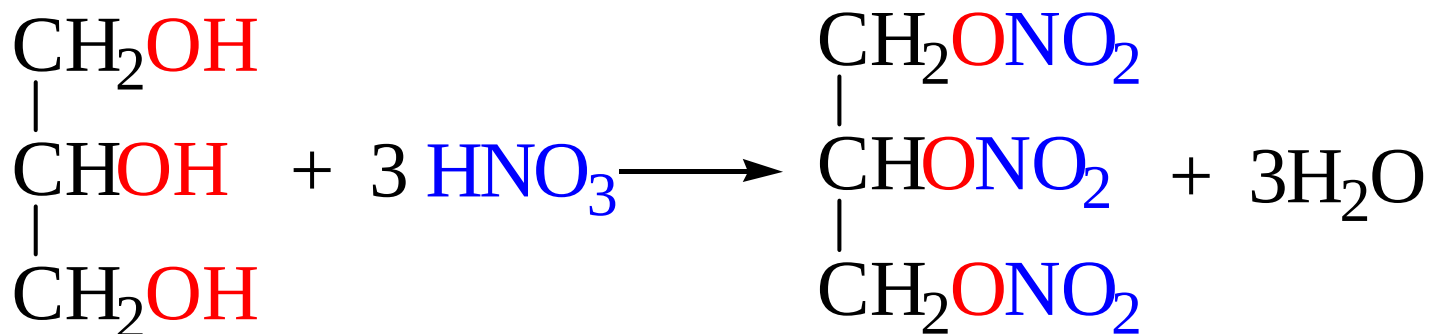


乳化剂



乙二醇

乙二醇二硝酸酯



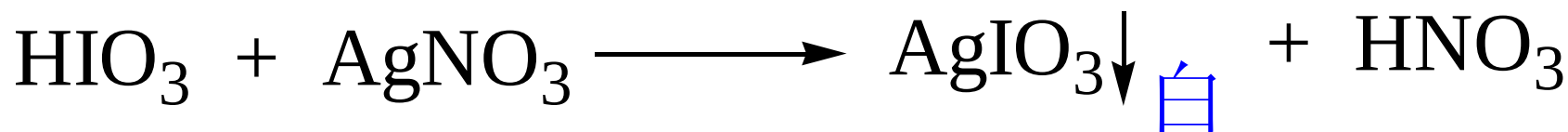
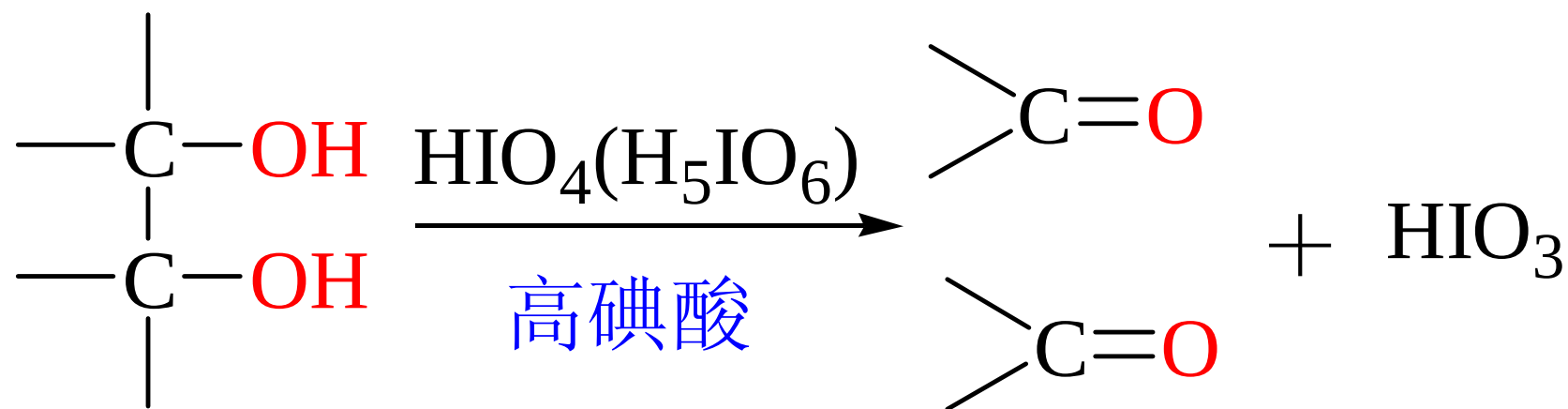
甘油

甘油三硝酸酯
硝化甘油

烈性炸药

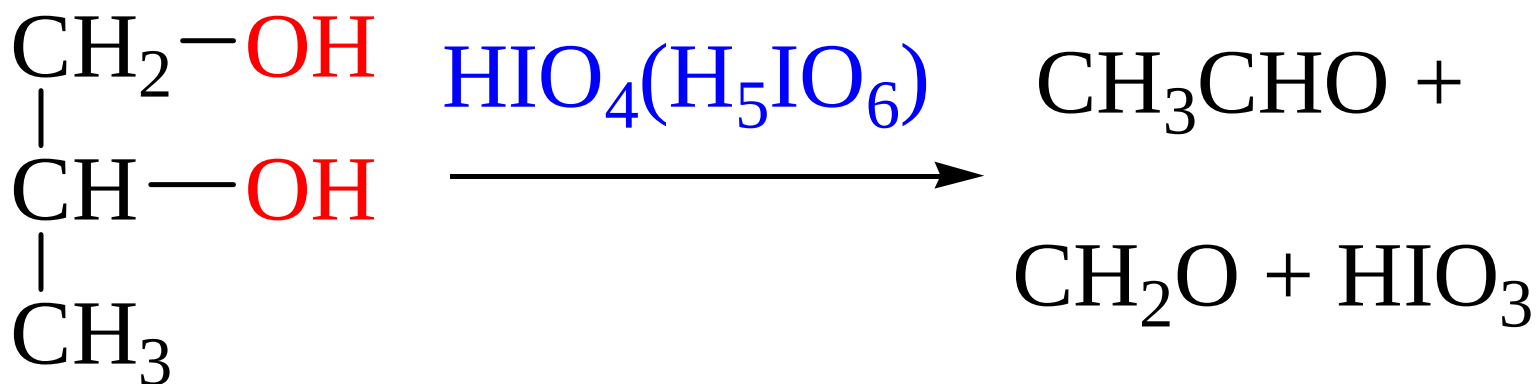
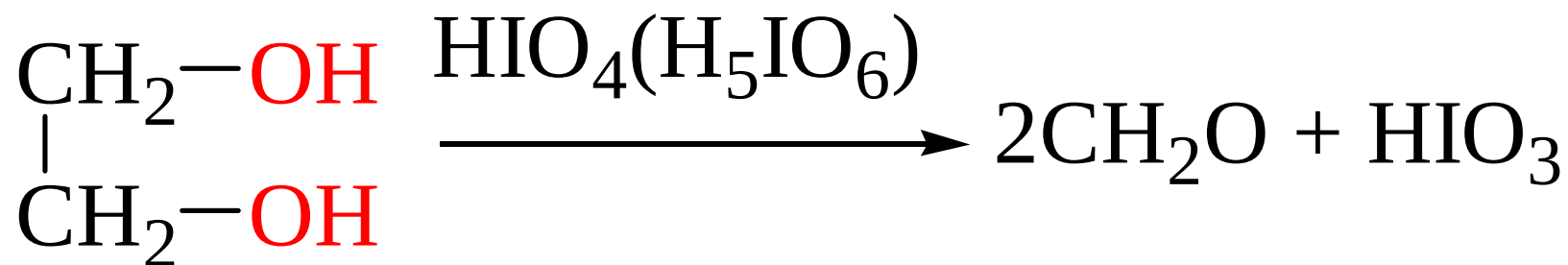
9.3.6 邻二醇的反应

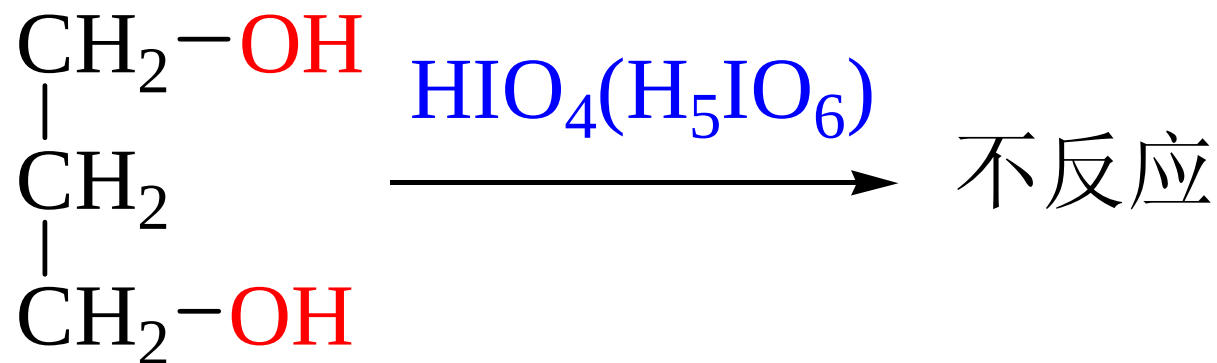
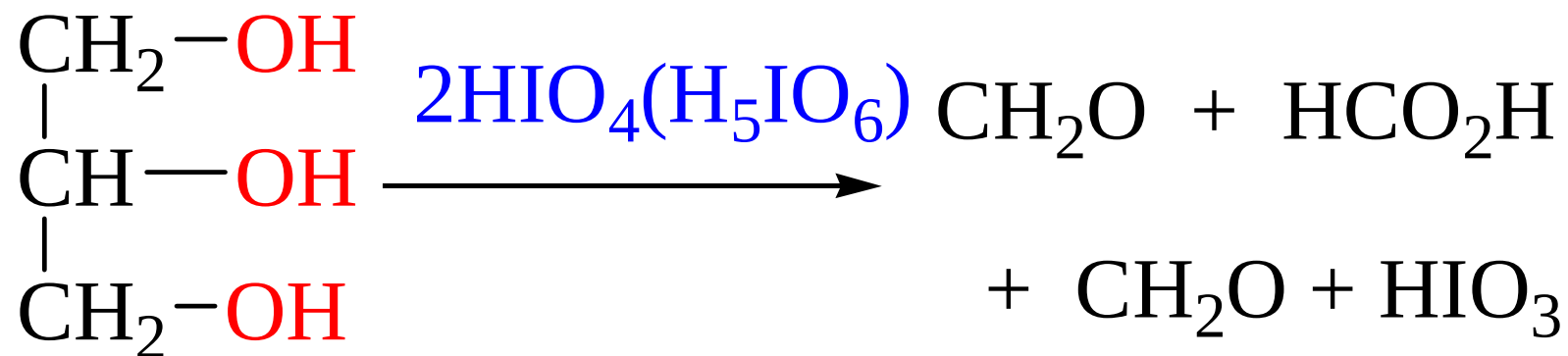
1. 被高碘酸氧化



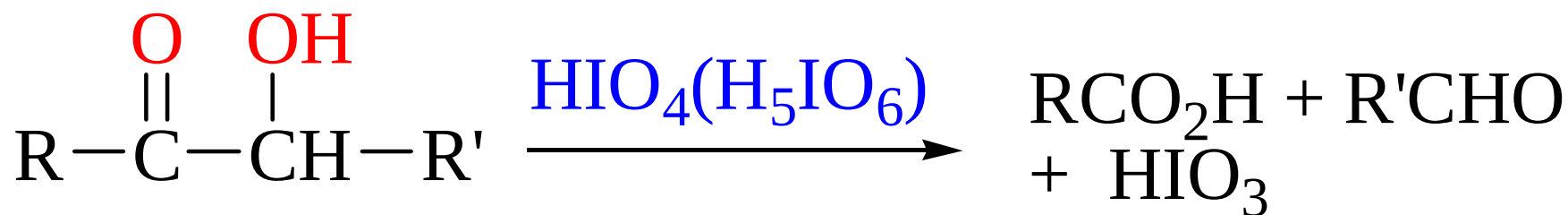
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Example





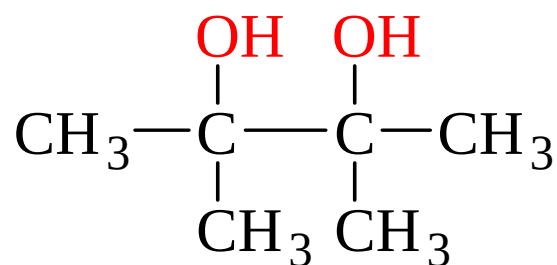
Example



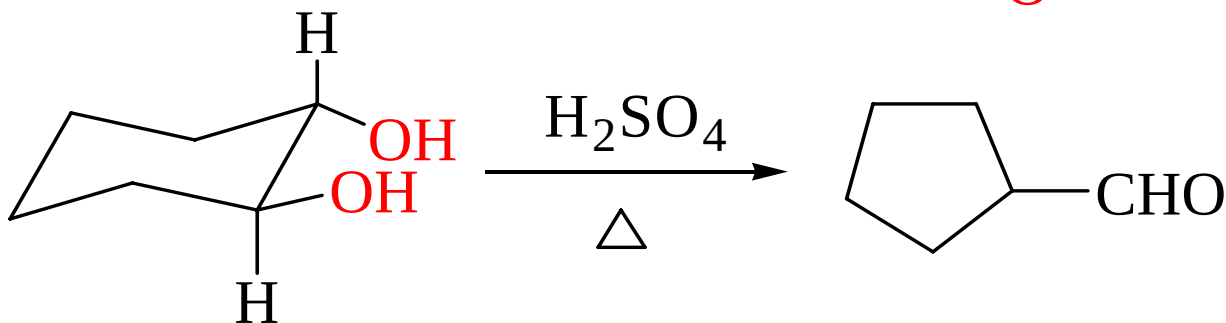
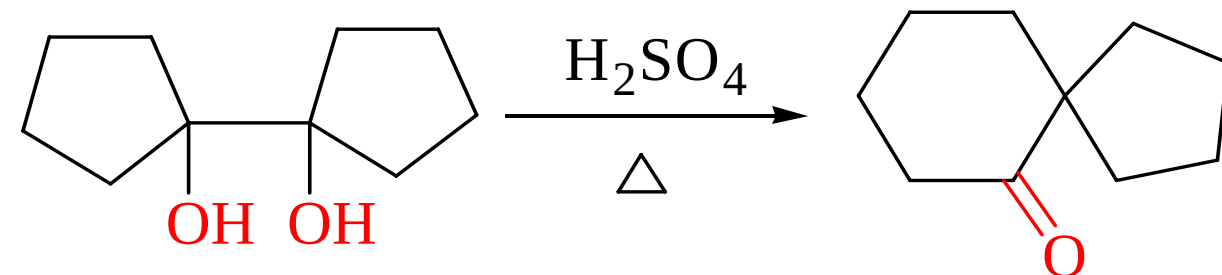
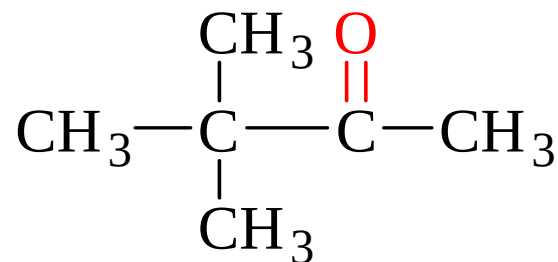
9.3.6 邻二醇的反应

2. 频哪醇重排

Pinacol

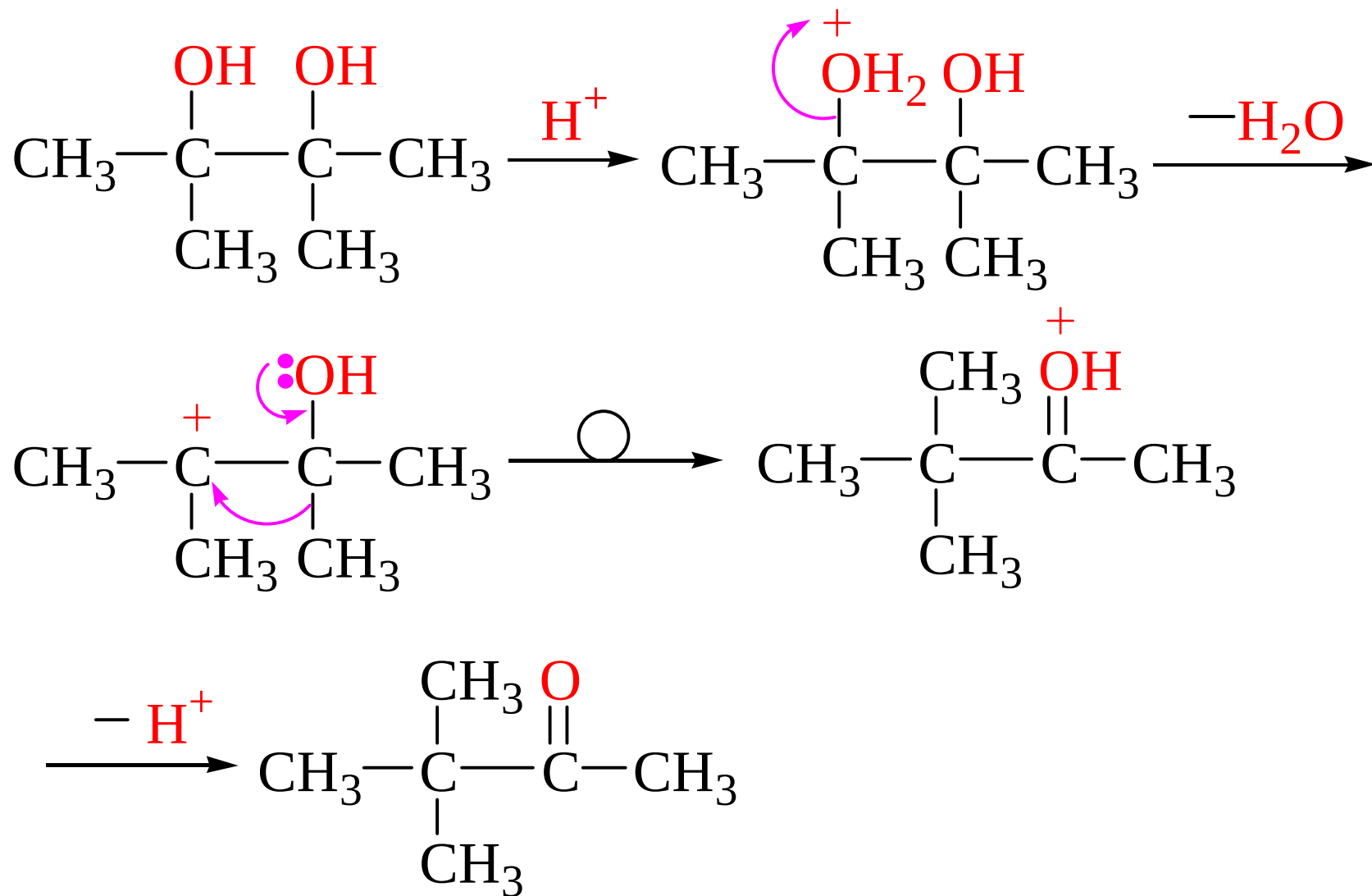


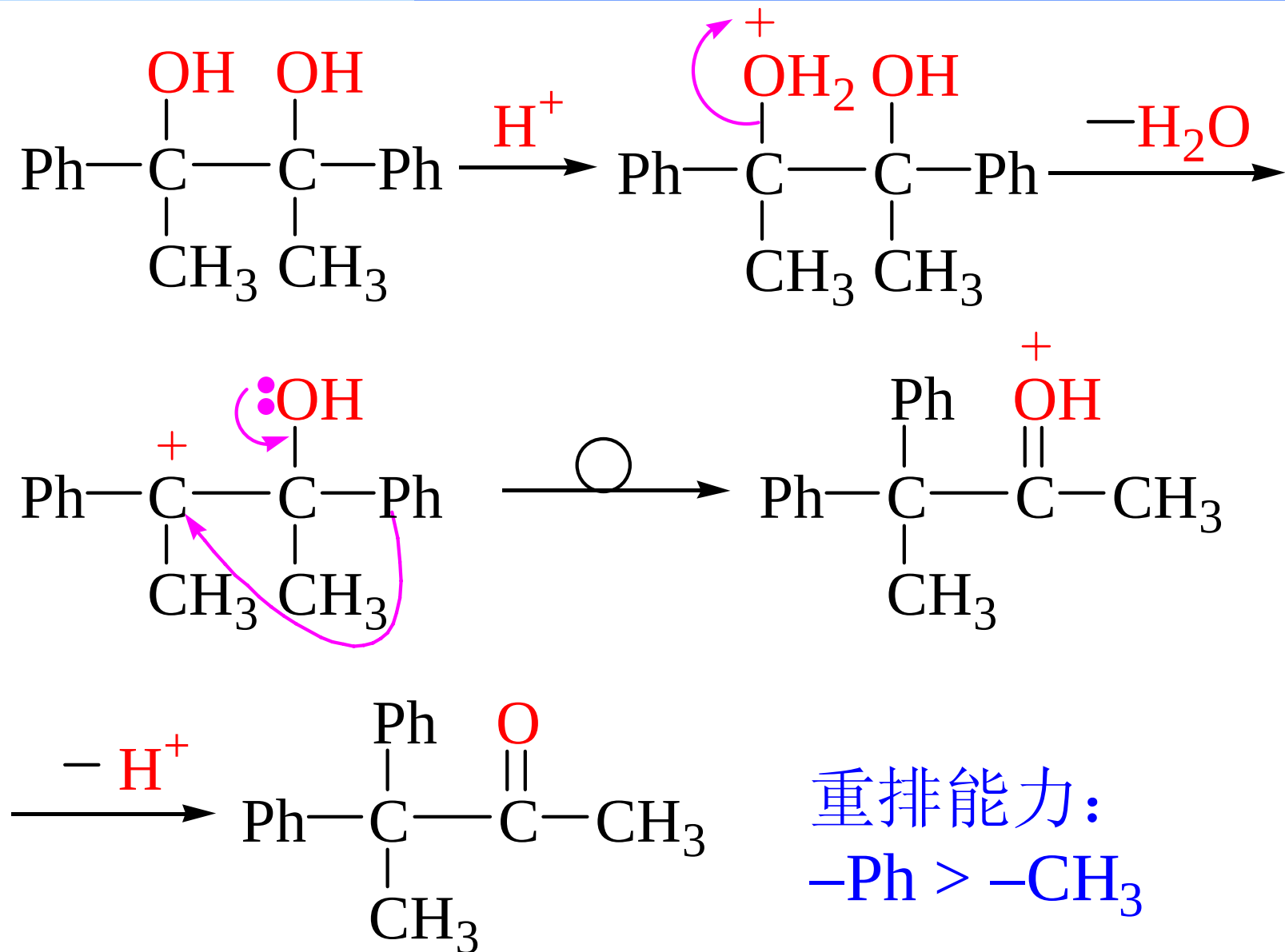
Pinacolone



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反应机理



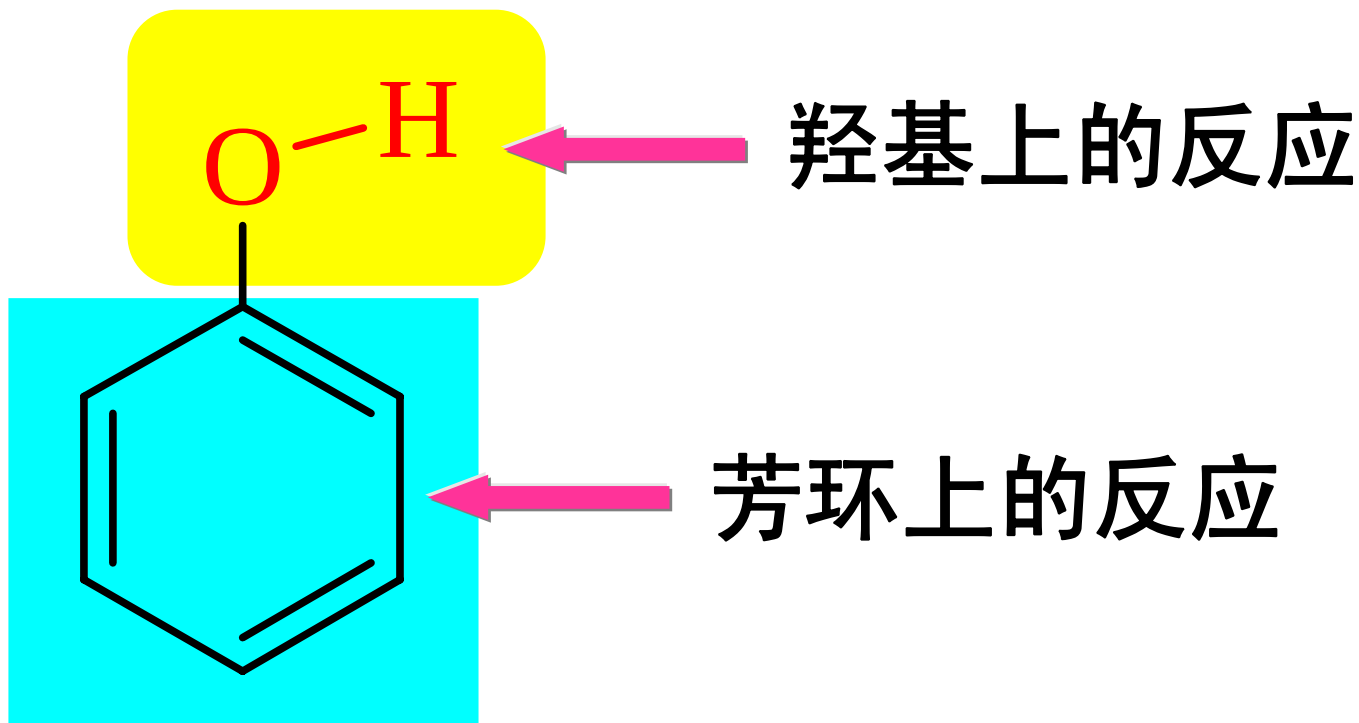


9.4 酚的结构、命名、来源

和物理性质(自学)

- 分类：根据分子中羟基的数目分为一元酚，二元酚和多元酚
- 命名：一般以苯酚为母体命名
- 物理性质：具有特殊气味，能形成分子间氢键，沸点较高，水中有一定溶解度

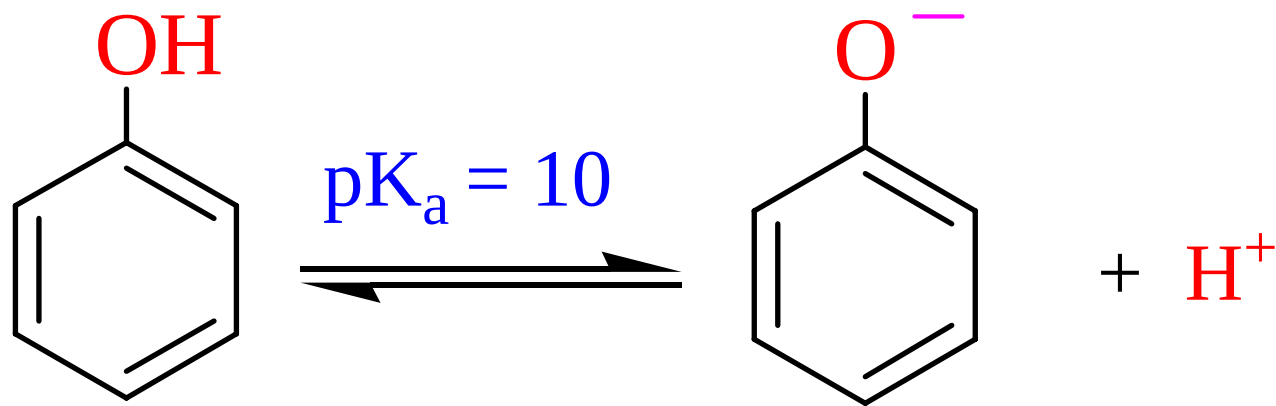
9.5 酚的化学性质



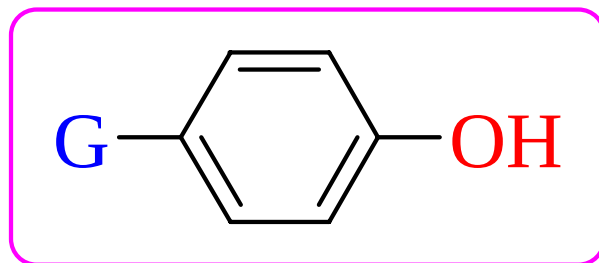
厦门大学有机化学B

9.5.1 羟基上的反应

1. 酚的酸性

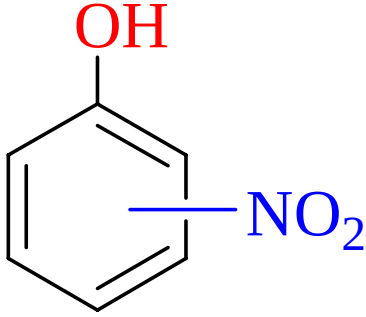


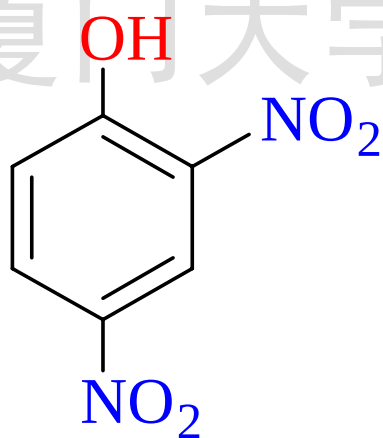
取代酚的酸性



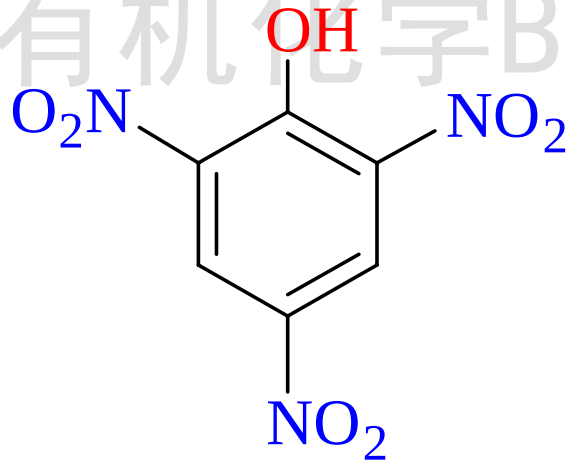
G	NH ₂	CH ₃ O	CH ₃	H	Cl	Br	I	NO ₂
pK _a	10.46	10.21	10.17	10.0	9.38	9.35	9.30	7.15



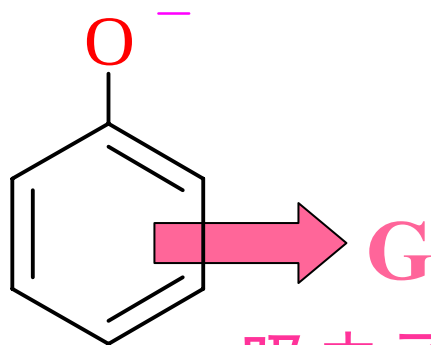
	NO ₂	<i>m</i> -	<i>o</i> -	<i>p</i> -
pK _a	8.28	7.17	7.15	



pK_a 4.09



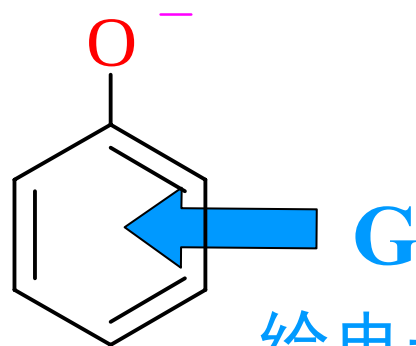
0.25



吸电子取代基

负离子稳定性提高

酸性增强



给电子取代基

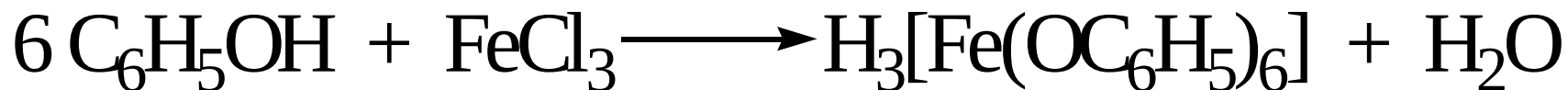
负离子稳定性降低

酸性减弱

厦门大学有机化学B

9.5.1 羟基上的反应

2. 酚与三氯化铁的显色反应



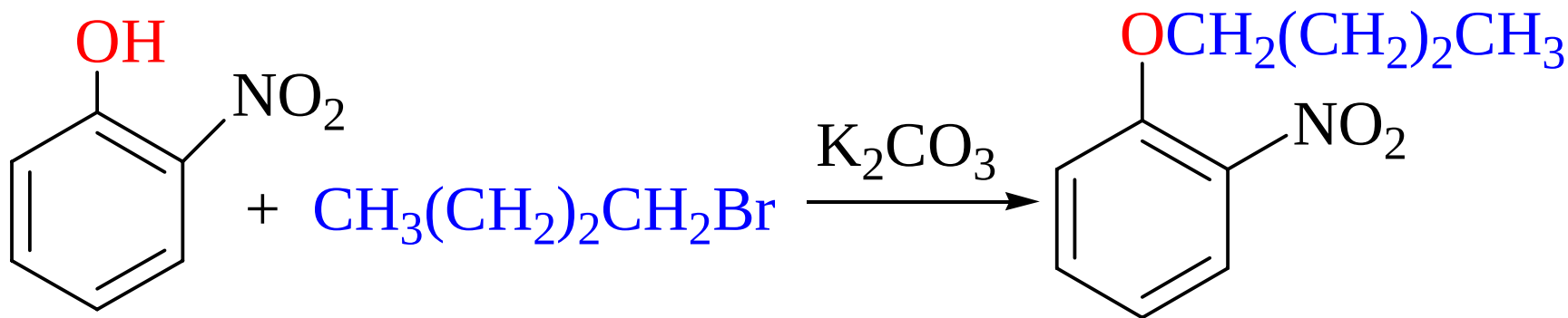
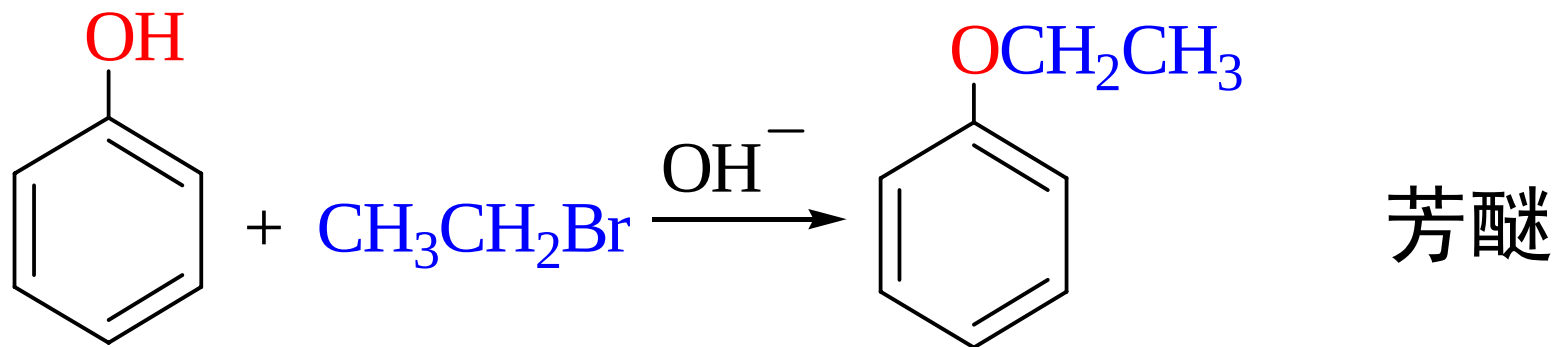
蓝紫色

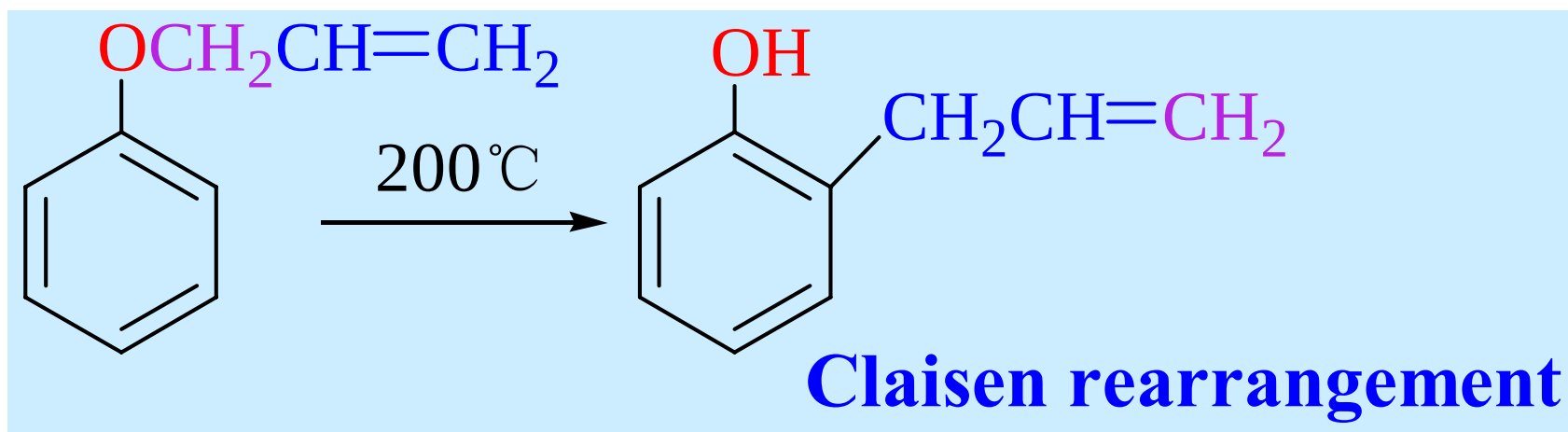
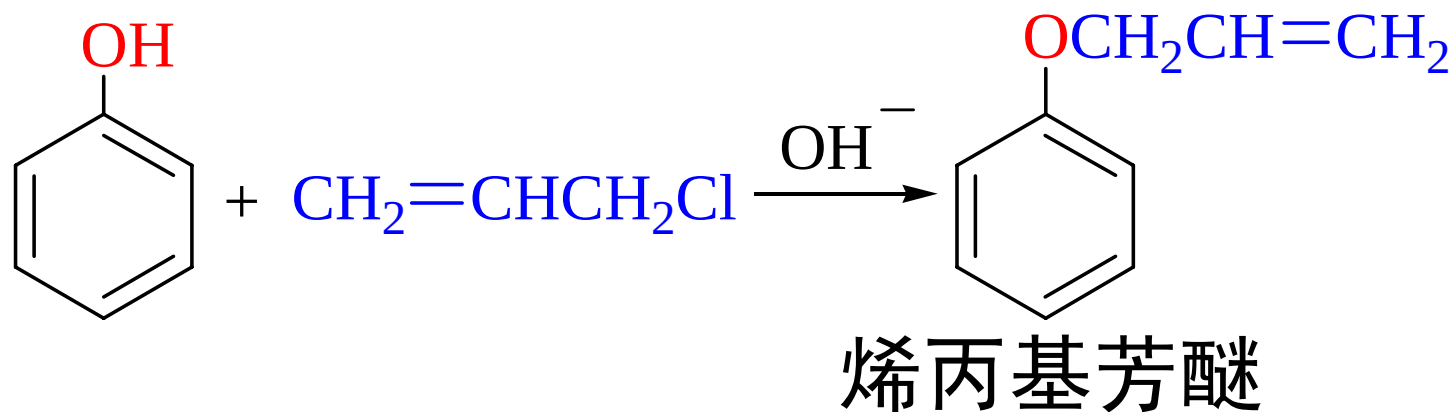
烯醇式结构的化合物大多能使三氯化铁显色

厦门大学有机化学B

9.5.1 羟基上的反应

3. 烷基化反应和Claisen 重排

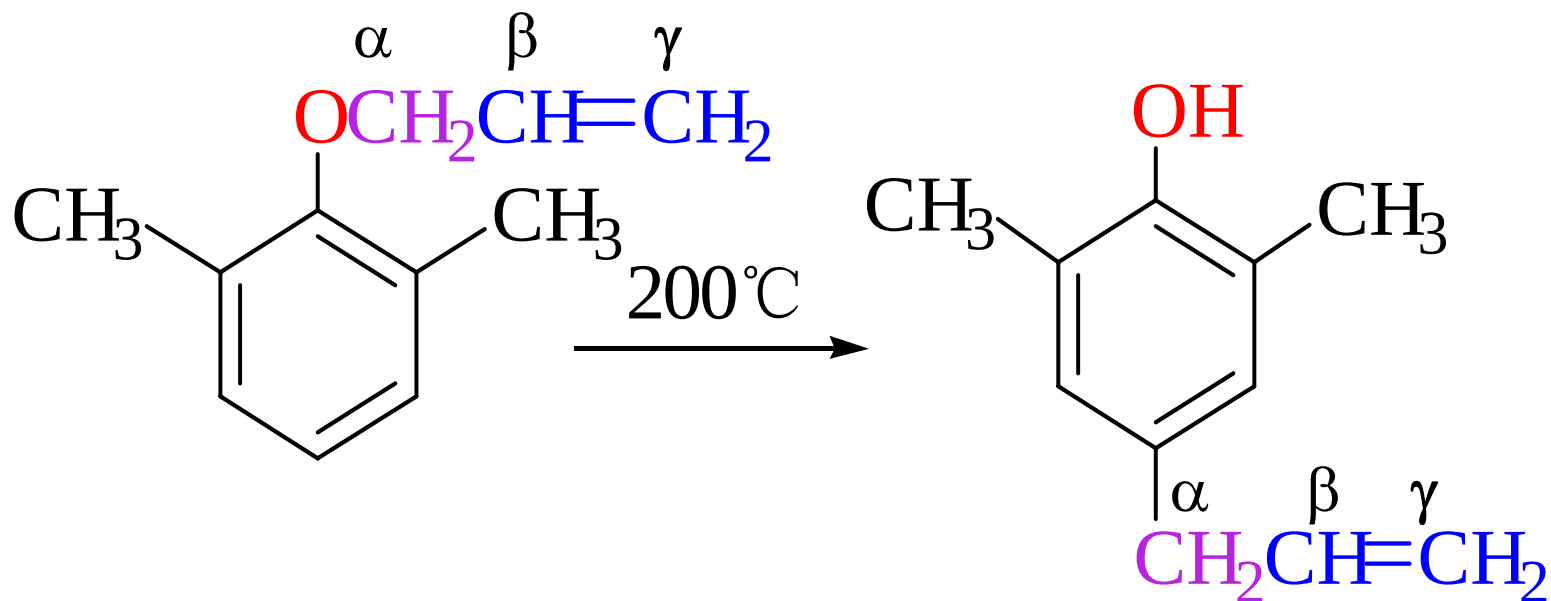
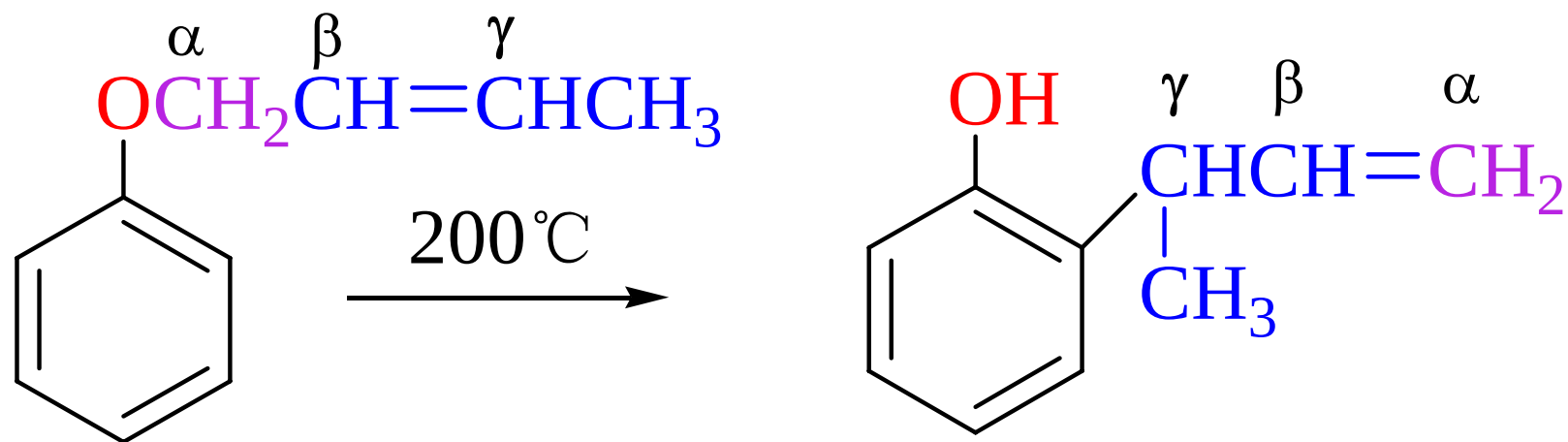


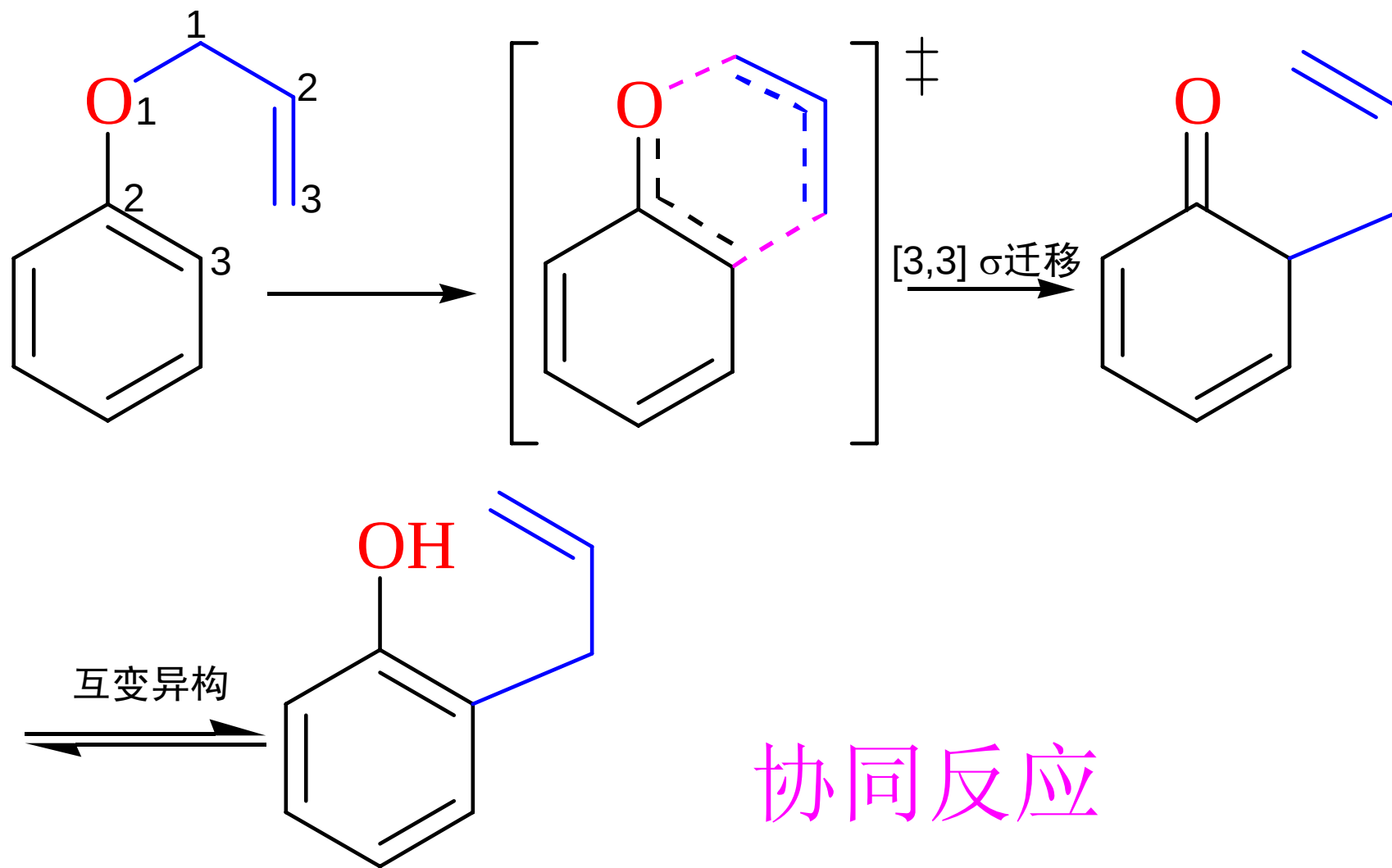


克莱森重排

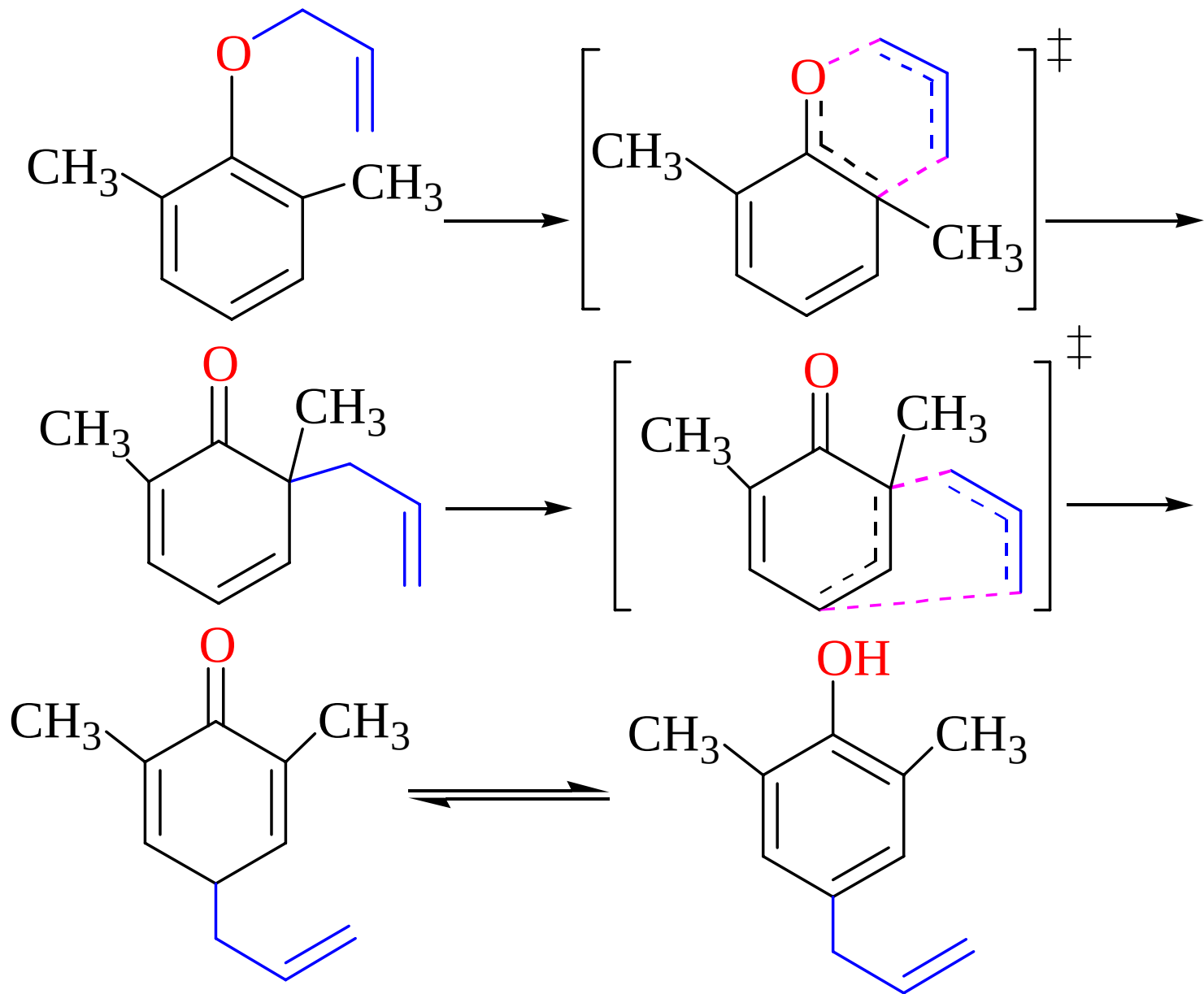
厦门大学有机化学B

Example





厦门大学 反应机理 化学B

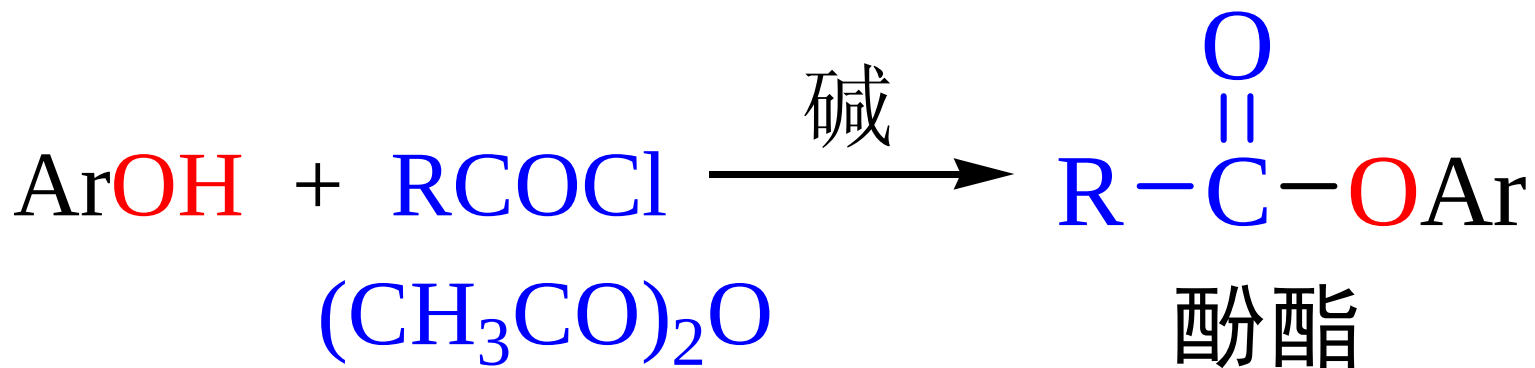


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9.5.1 羟基上的反应

4. 酰基化反应及Fries重排

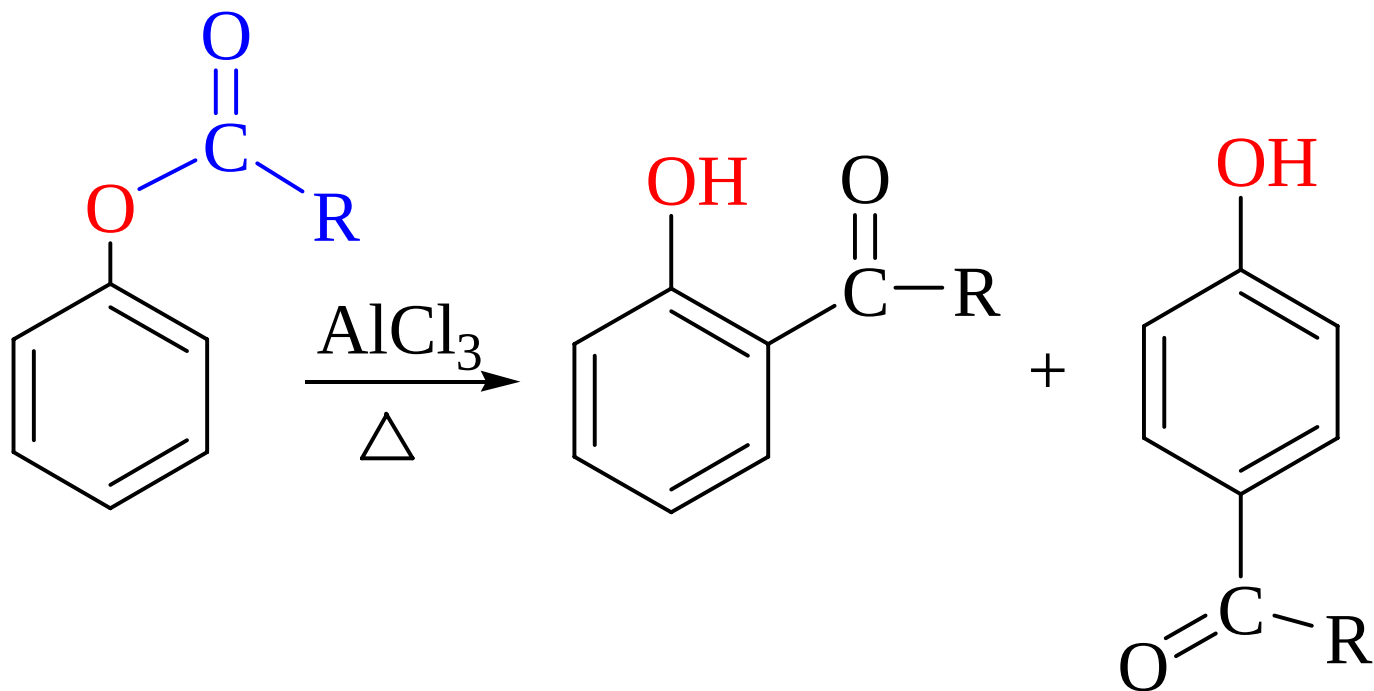
一般用酰氯或酸酐反应



碱：吡啶，NaOH，Na₂CO₃，NaOAc

厦门大学有机化学B

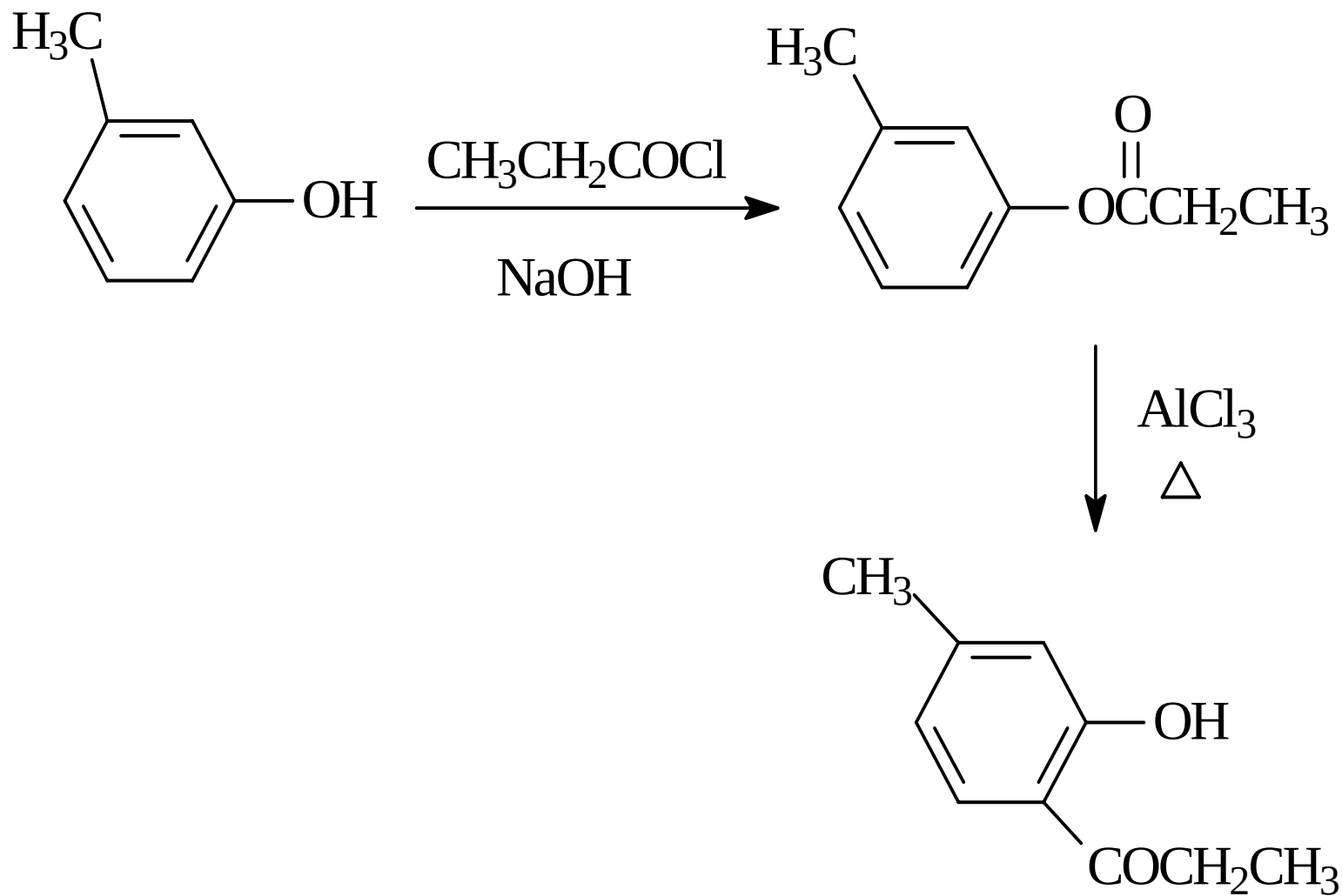
Fries重排



高温下产物以邻位为主
较低温度下产物以对位为主

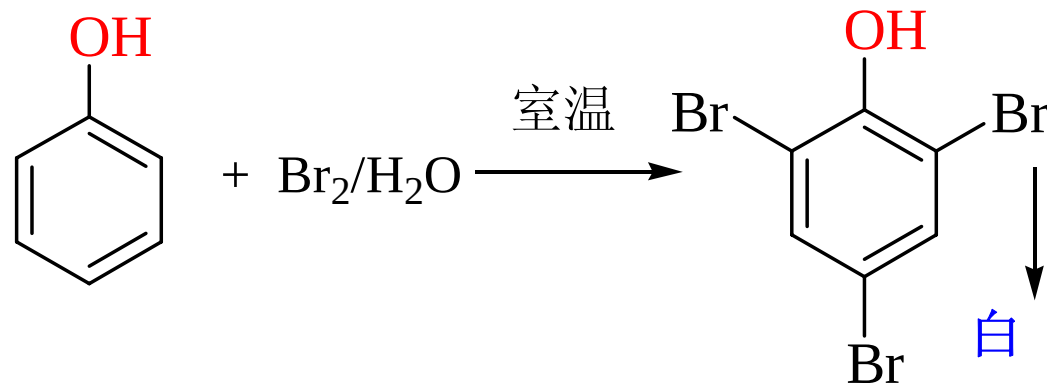
厦门大学有机化学B

Example

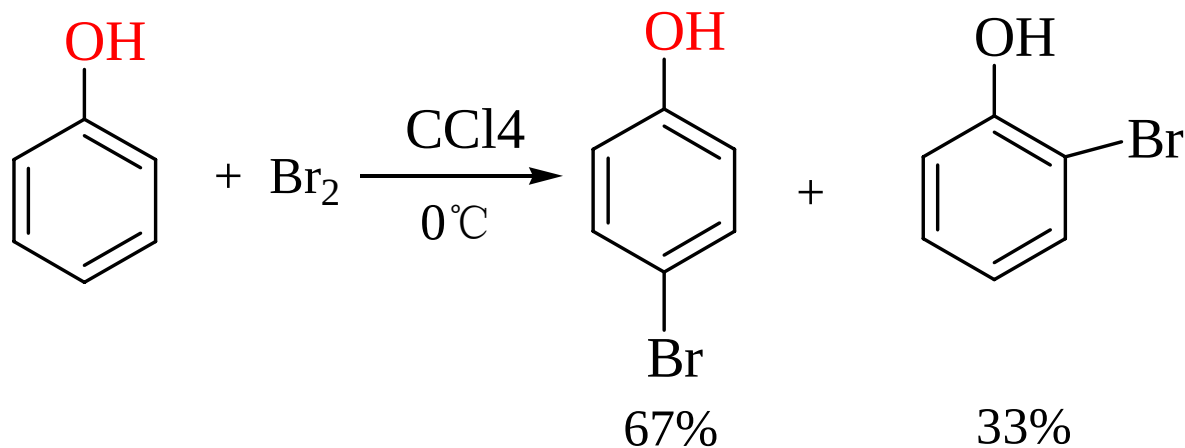


9.5.2 芳环上的取代反应(自学)

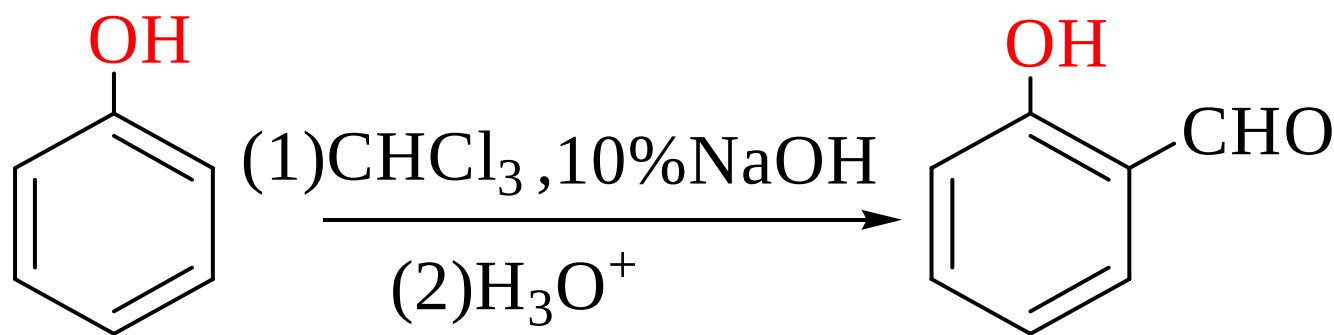
卤化反应



可用于鉴别苯酚



赖默-梯曼反应

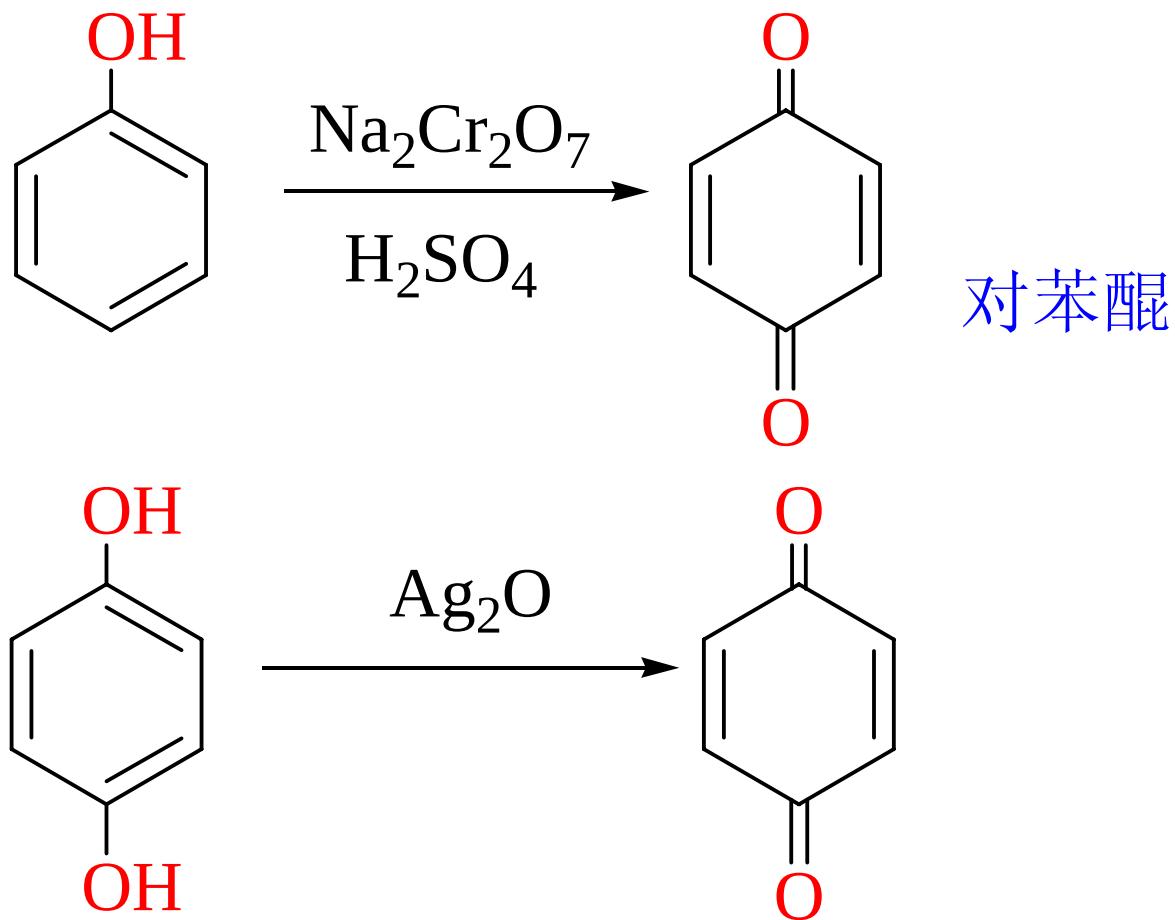


水杨醛

Reimer-Tiemann reaction

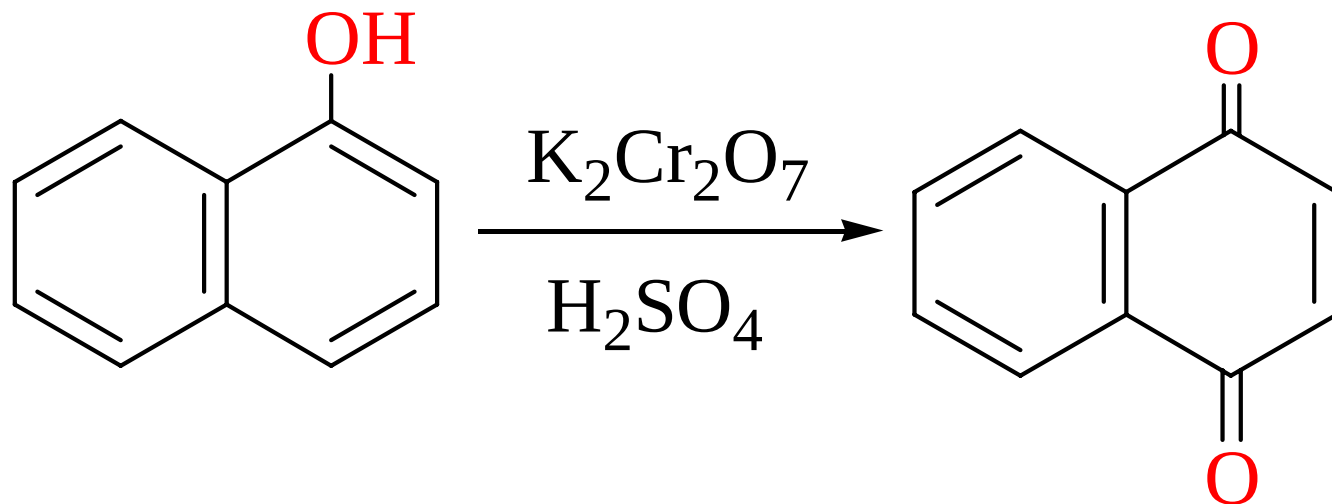
厦门大学有机化学B

9.5.3 酚的氧化反应

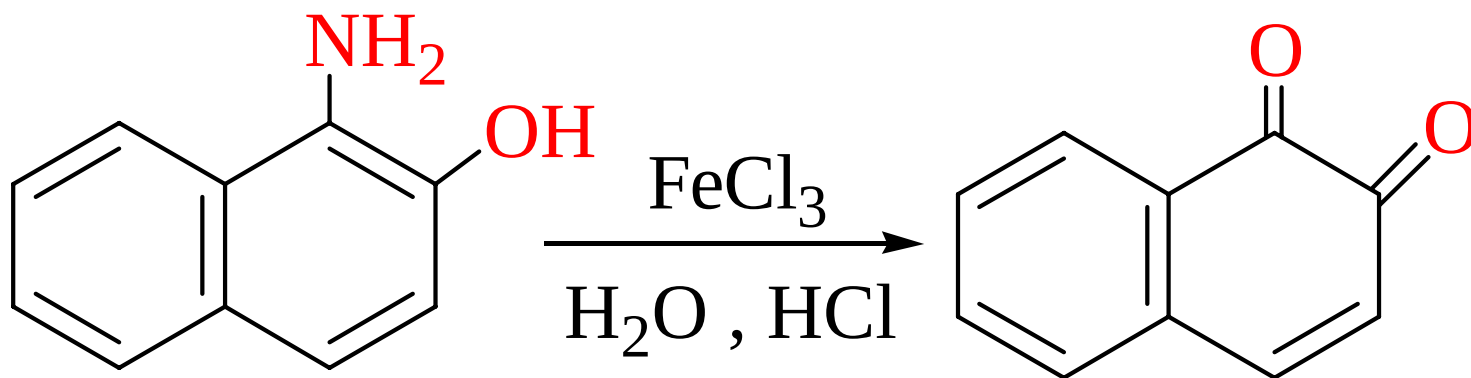


厦门大学有机化学B

Example

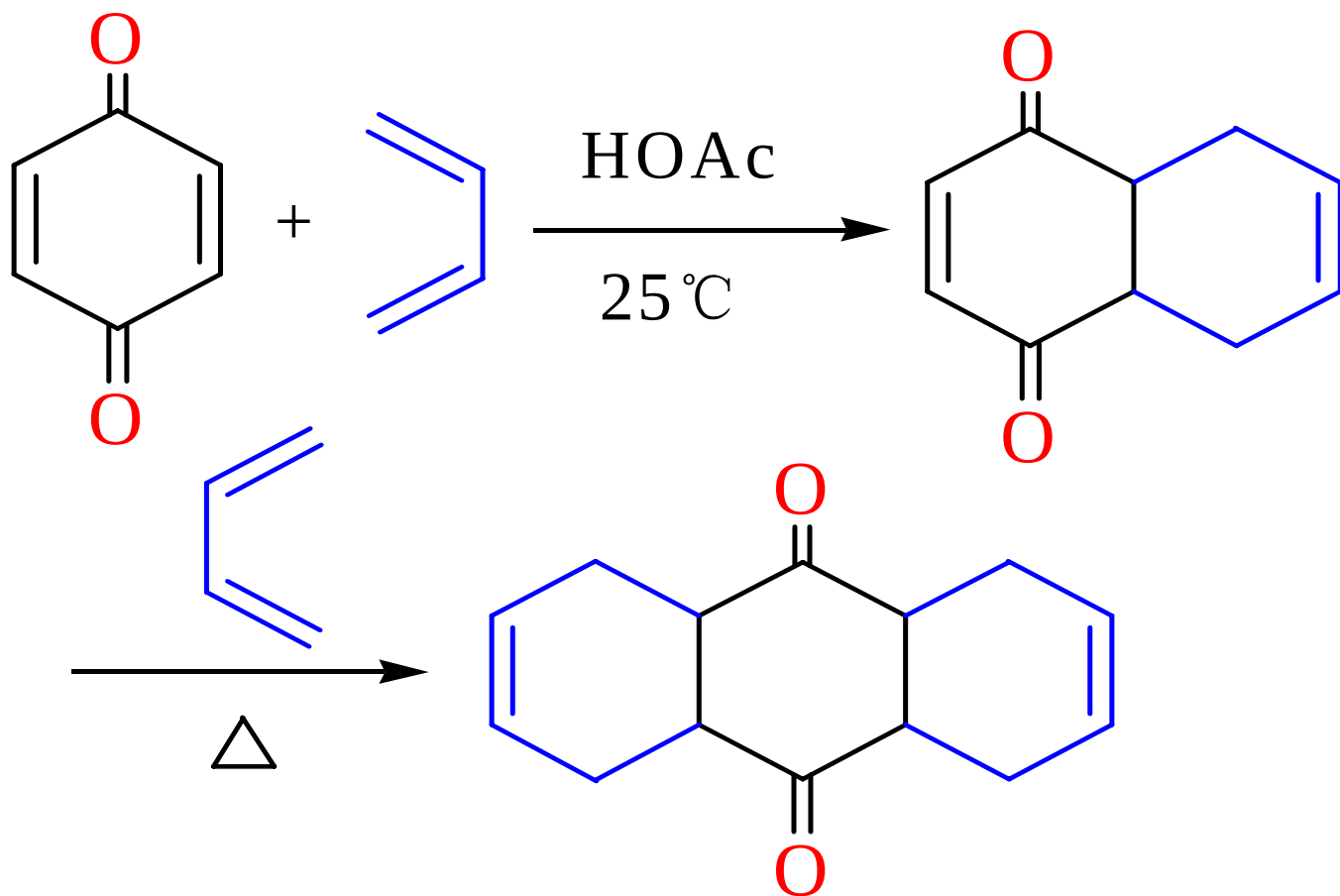


1,4-萘醌



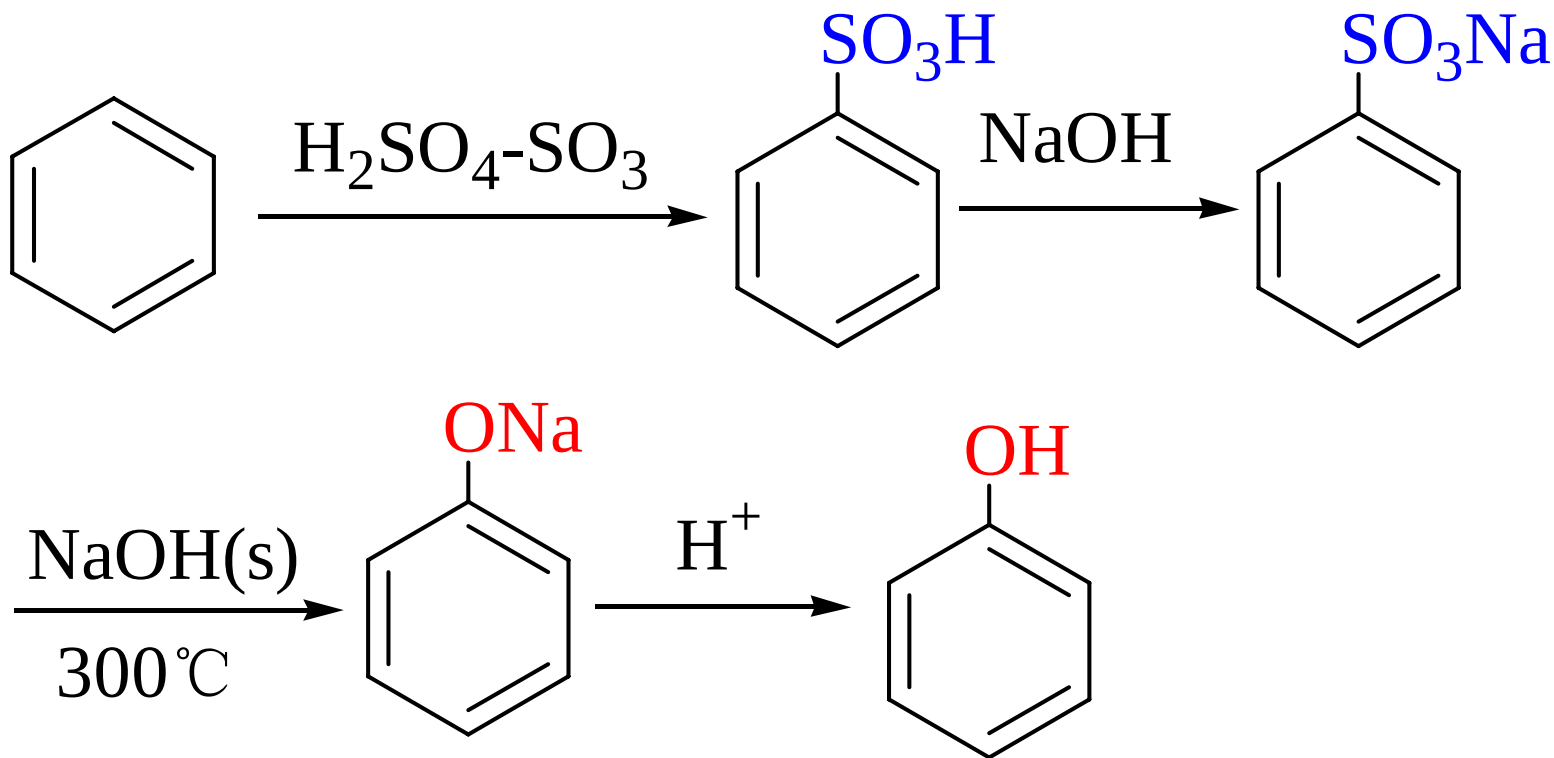
1,2-萘醌

对苯醌可作为亲双烯体 发生Diels—Alder反应

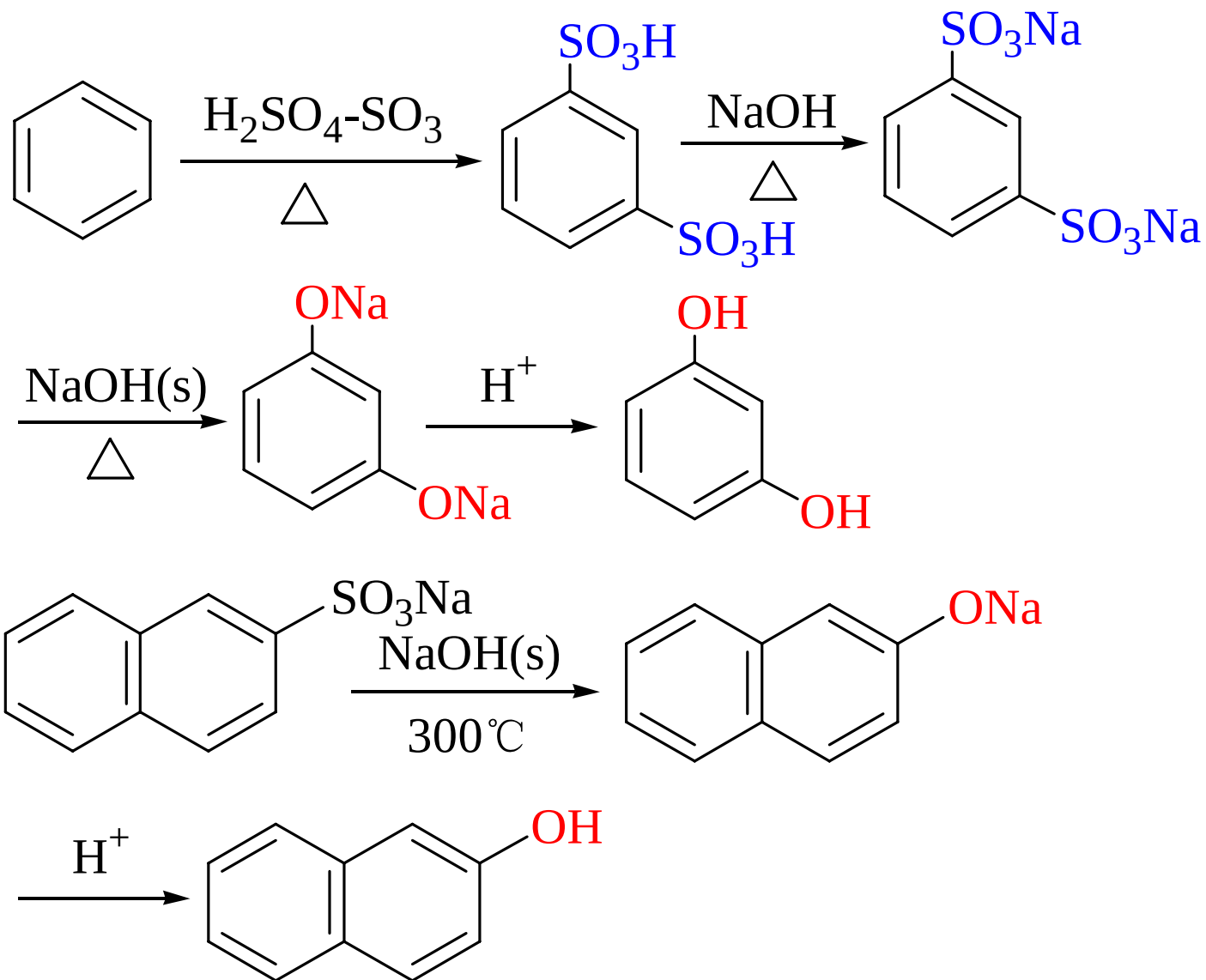


9.5.3 酚的制备

1. 磺酸盐碱熔法

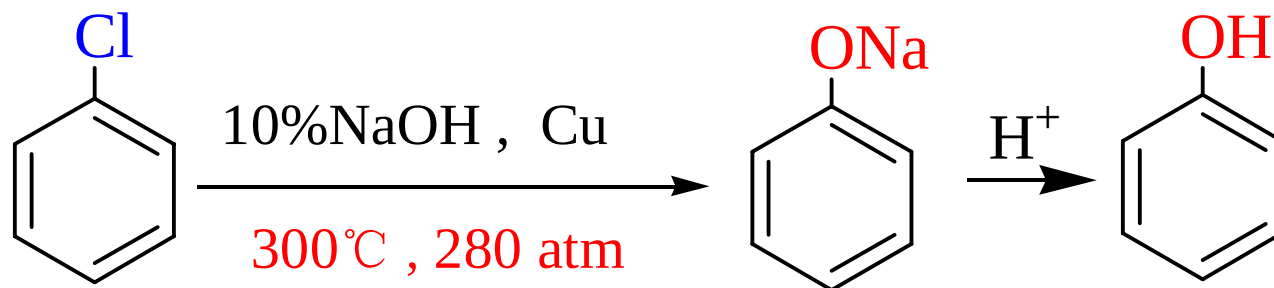


Example

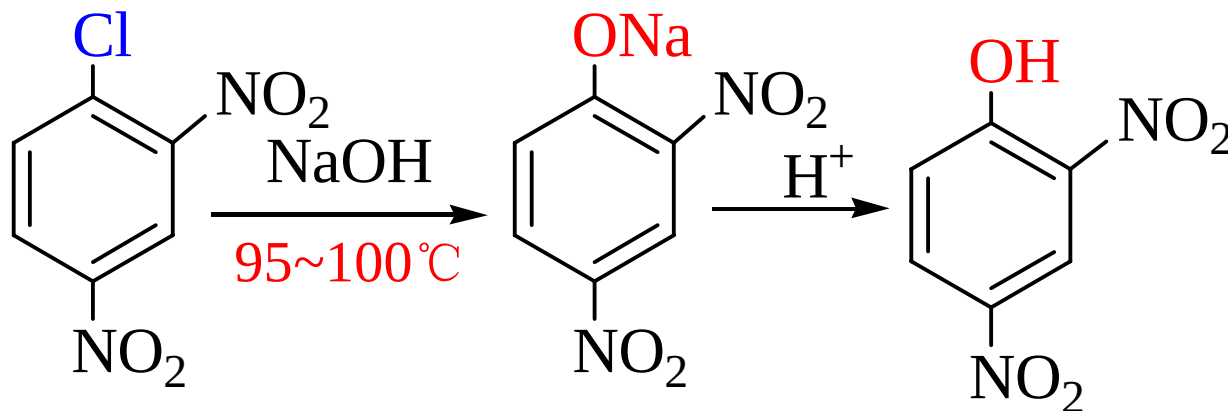


9.5.3 酚的制备

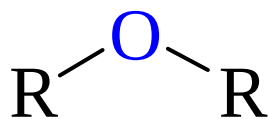
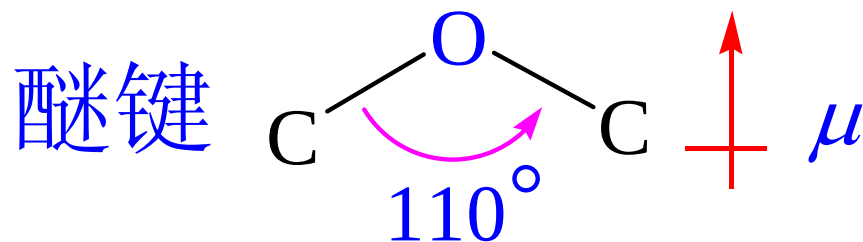
2. 氯苯水解法



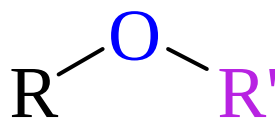
卤素的邻、对位有强吸电子基团时，水解变得容易进行：



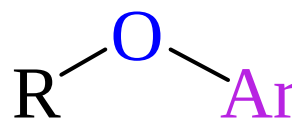
9.6 醚



简单醚
(对称醚)



混合醚
(不对称醚)

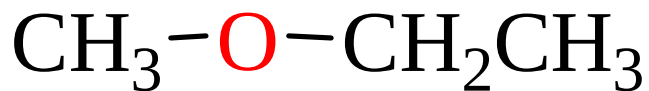


芳醚

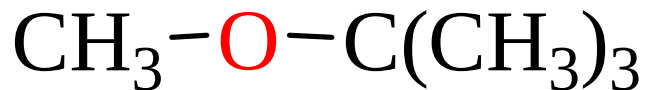
9.6.1 醚的命名

1. 无环醚

普通命名法：二个烃基名 + 醚
二个烃基名按“次序规则”列出

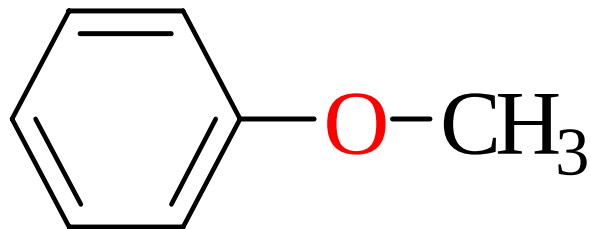


甲基乙基醚

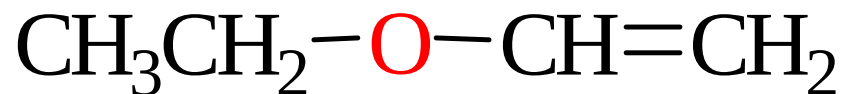


甲基叔丁基醚

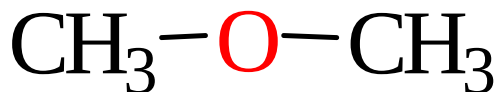
Example



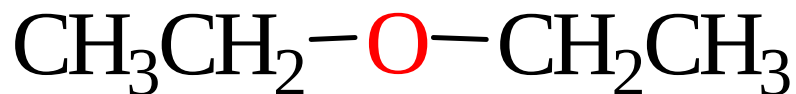
苯甲醚(茴香醚)



乙基乙烯基醚



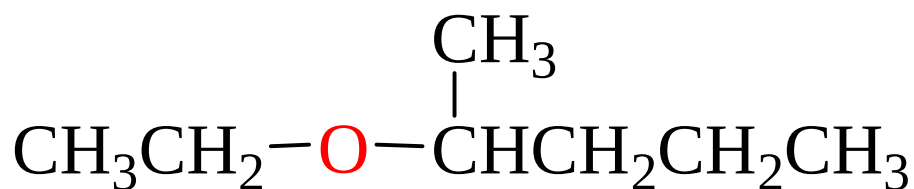
二甲醚(甲醚)



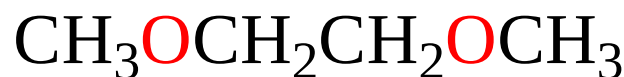
二乙醚(乙醚)

系统命名法：把醚看成是烃的烷氧衍生物，取较长的烃基做母体

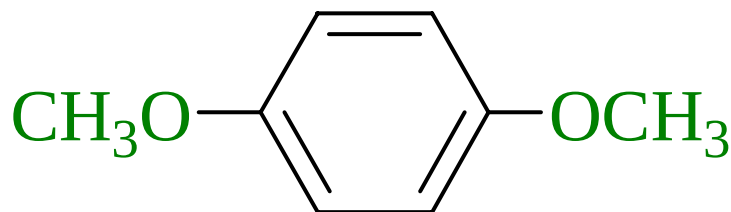
—OR 烷氧基



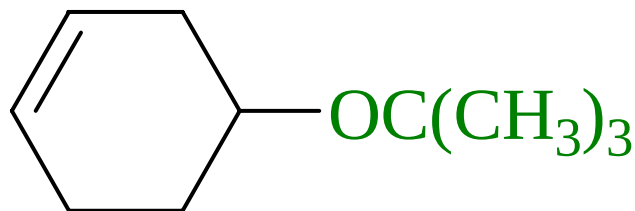
2-乙氧基戊烷



1,2-二甲氧基乙烷



1,4-二甲氧基苯



4-叔丁氧基-1-环己烯

9.6.1 醚的命名

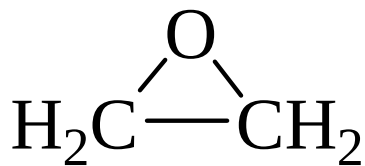
2. 环醚

环醚

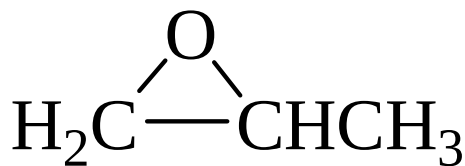
环氧化合物

大环多醚

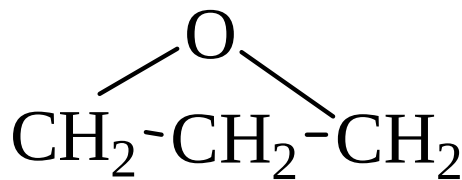
环氧化合物



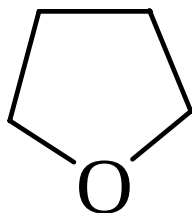
环氧乙烷



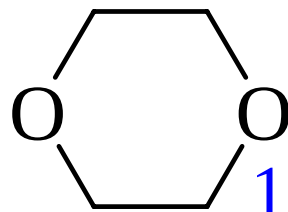
环氧丙烷



1,3-环氧丙烷



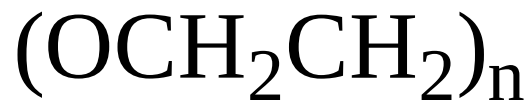
四氢呋喃
(THF)



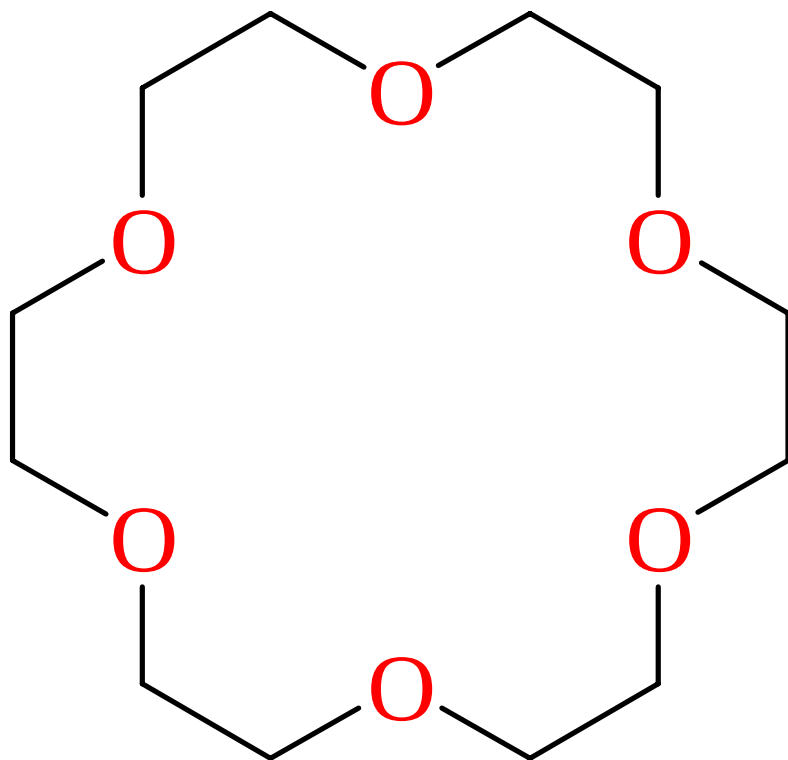
1,4-二氧六环

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大环多醚

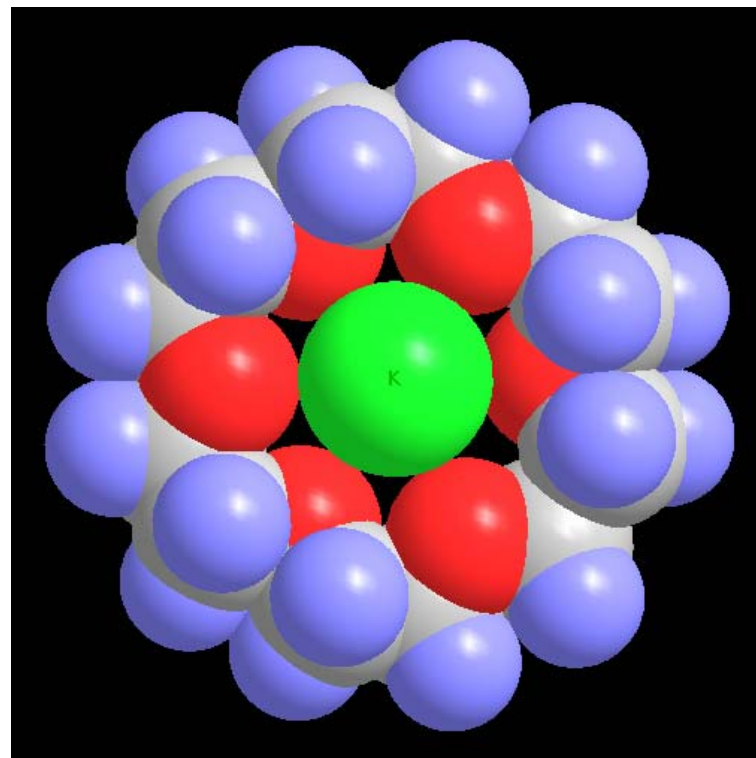
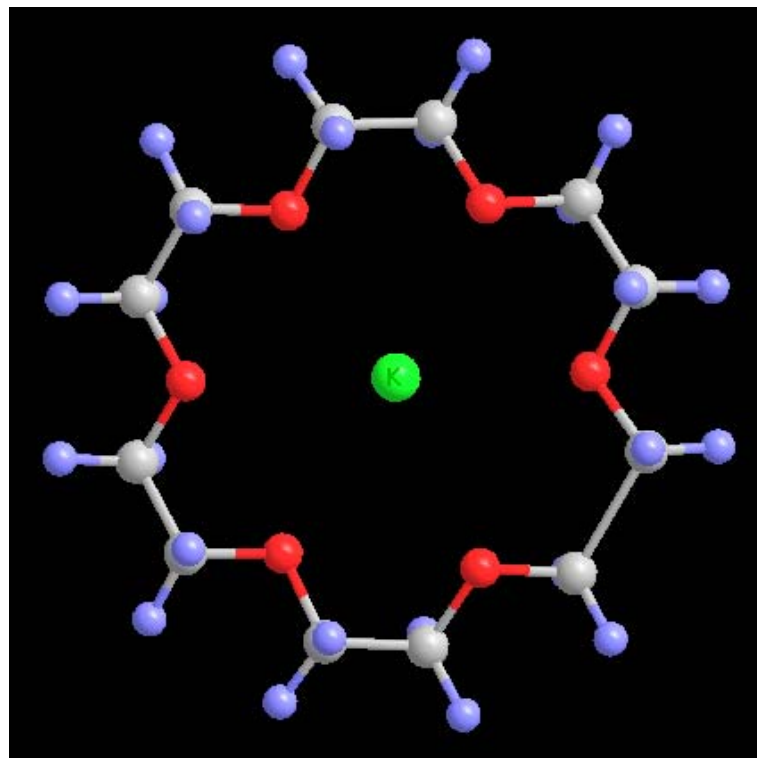
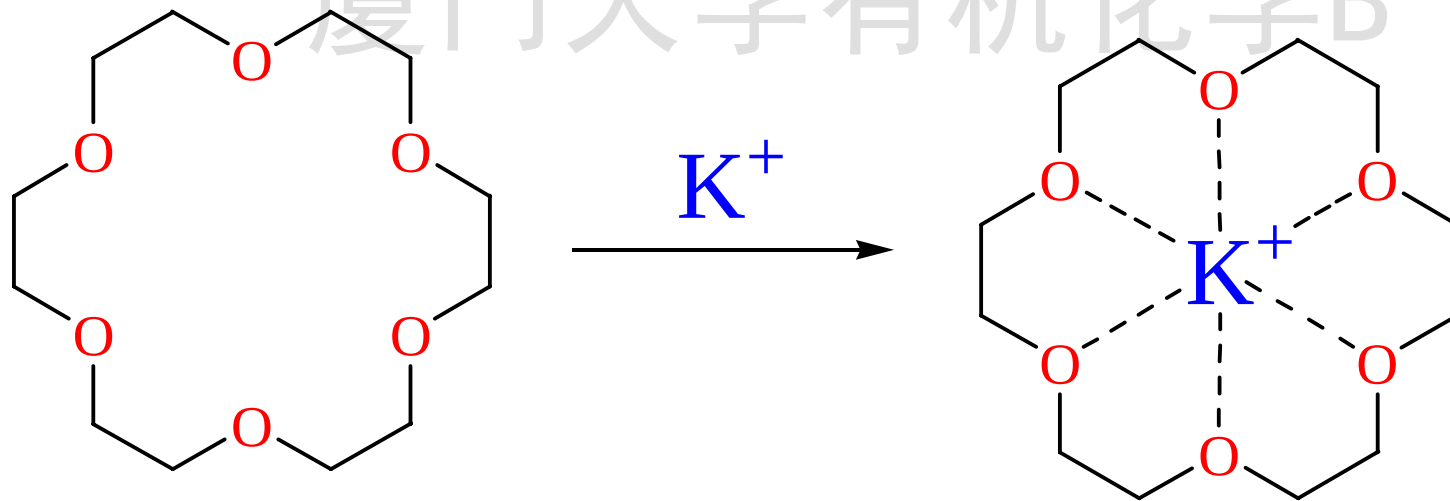


冠醚(Crown ether)



18—冠—6

18—crown—6



9.6.2 醚的物理性质

- b.p. 比同碳原子数的醇低得多
- 水中溶解度小(环醚除外)



b.p.(°C) -24.9

34.6

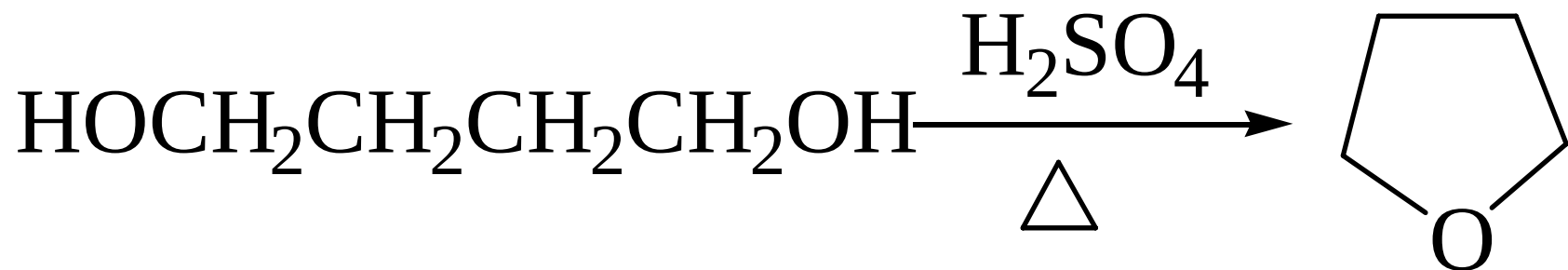
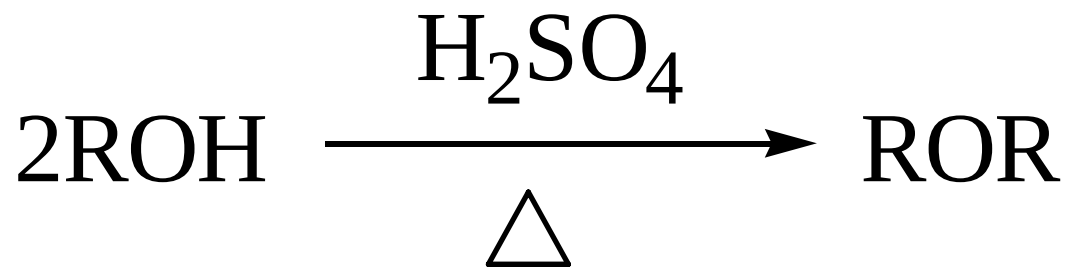


b.p.(°C) 78.4

117.8

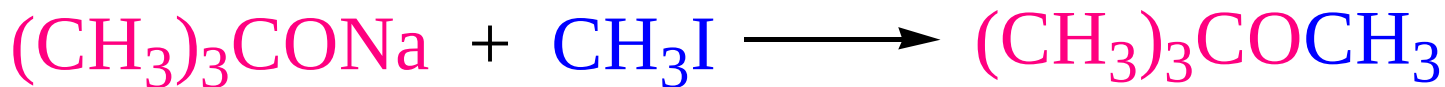
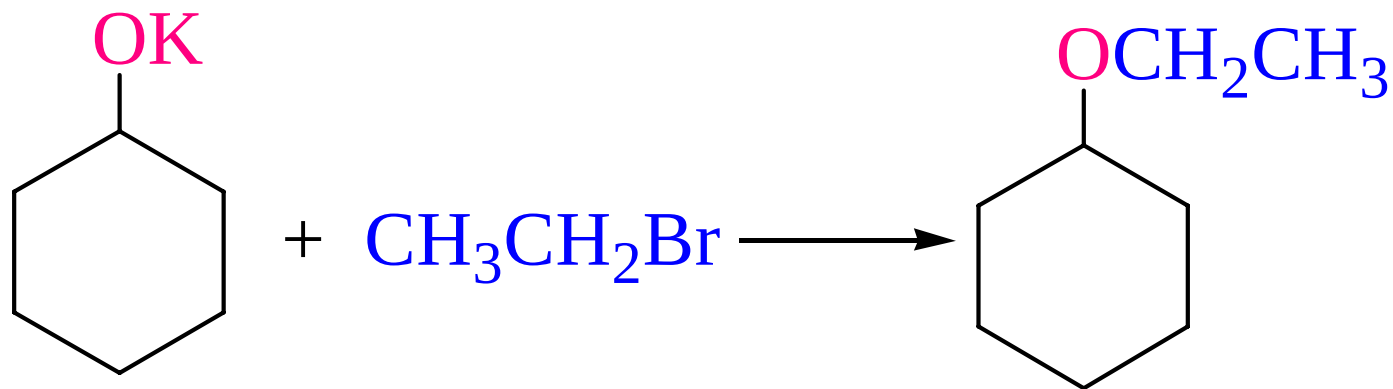
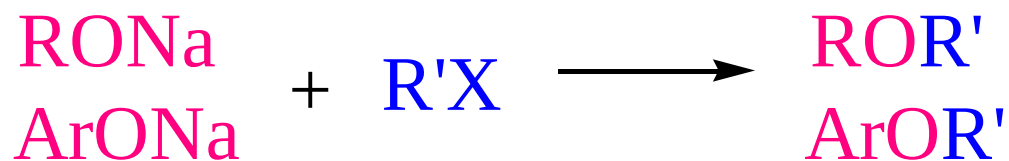
9.6.3 醚的制备

1. 醇的分子间脱水



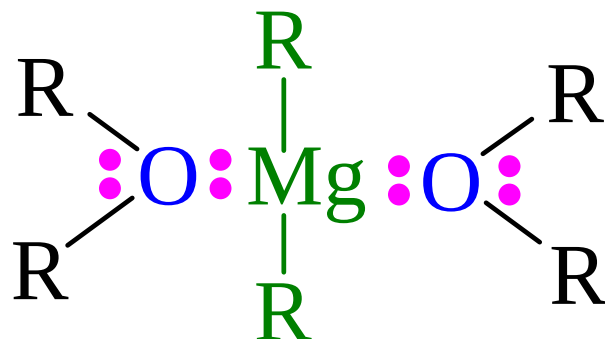
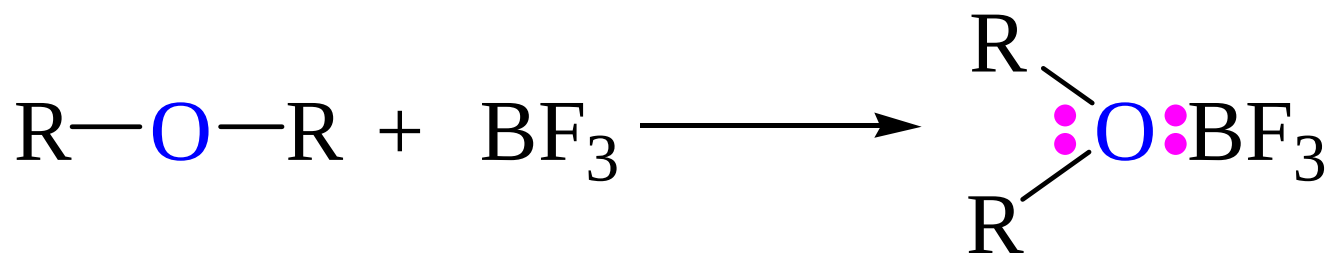
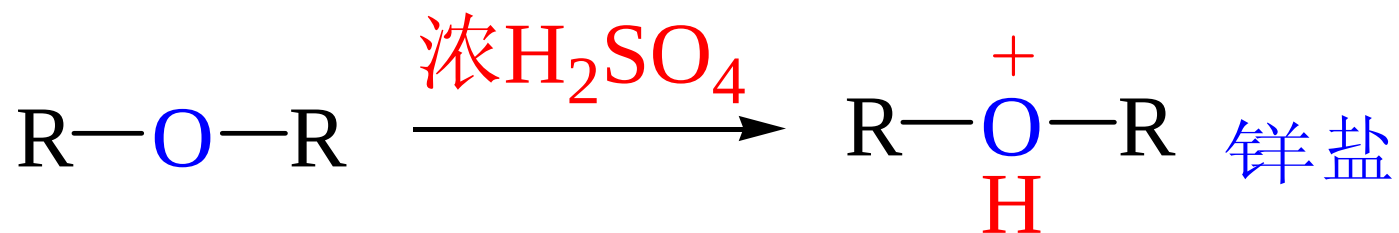
9.6.3 醚的制备

2. Williamson 合成法



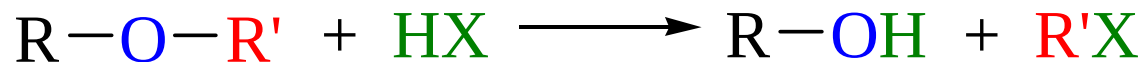
9.6.4 醚的反应

1. 形成 铯 盐

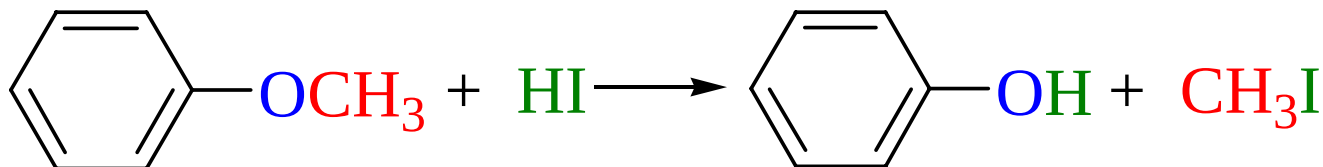
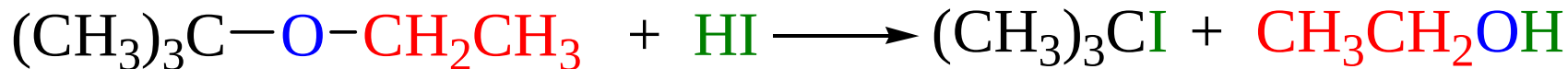
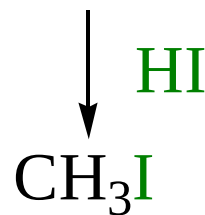
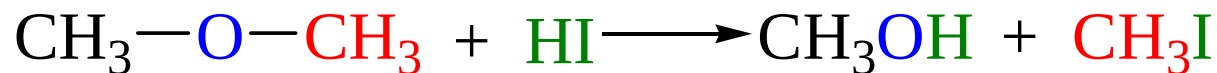


9.6.4 醚的反应

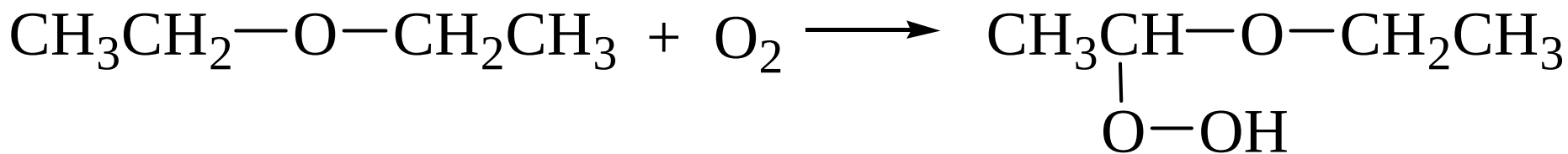
2. 醚键的断裂



反应活性 $\text{HI} > \text{HBr} > \text{HCl}$



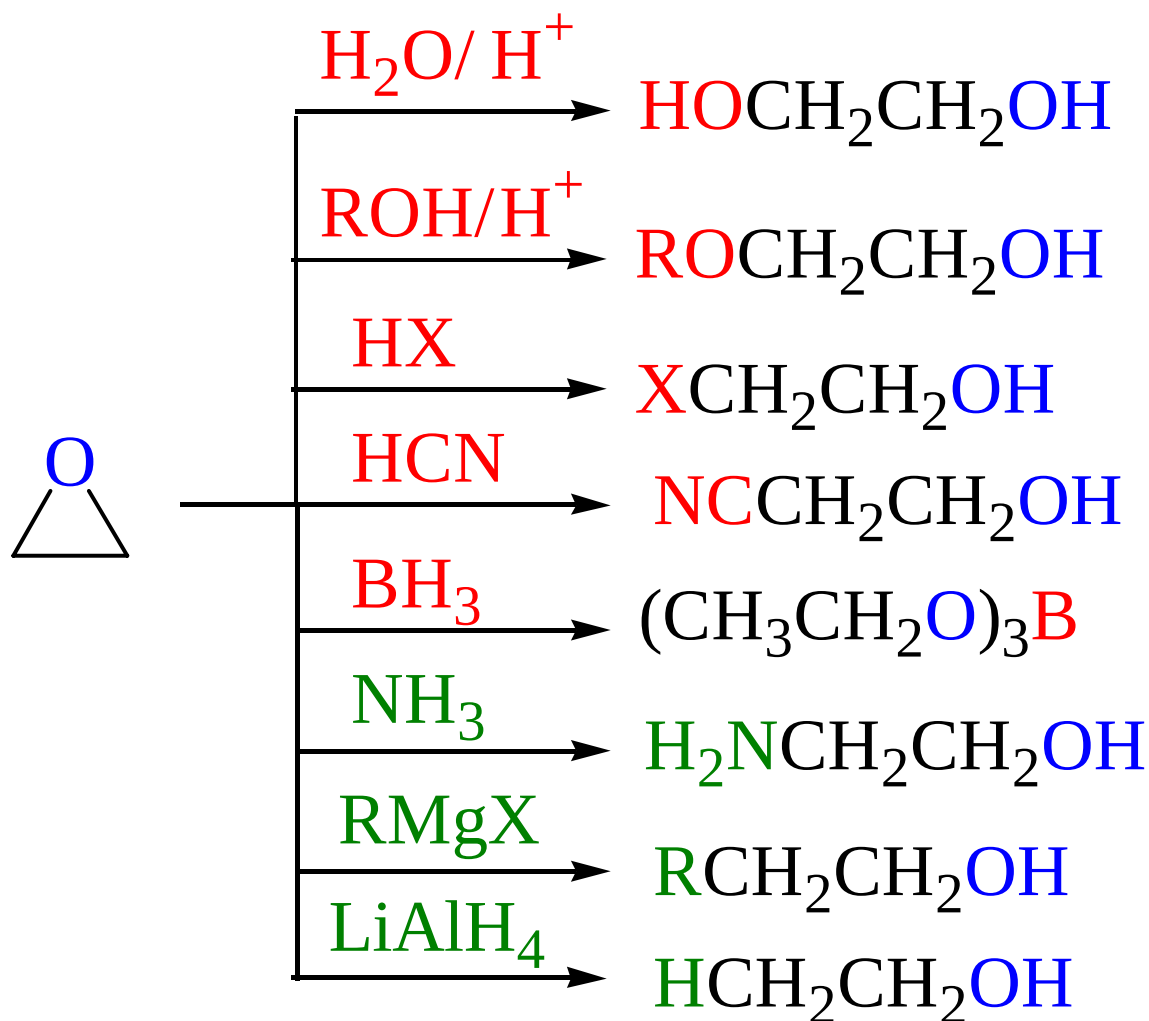
3. 过氧化物peroxide的生成



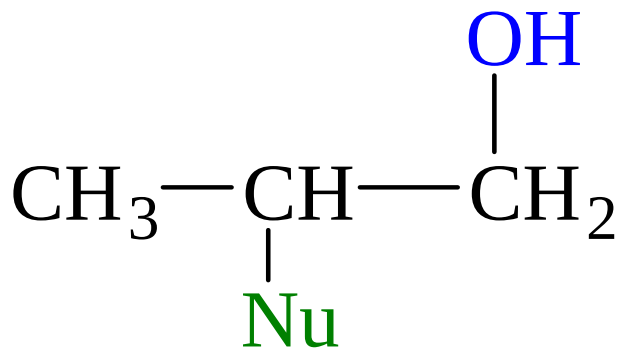
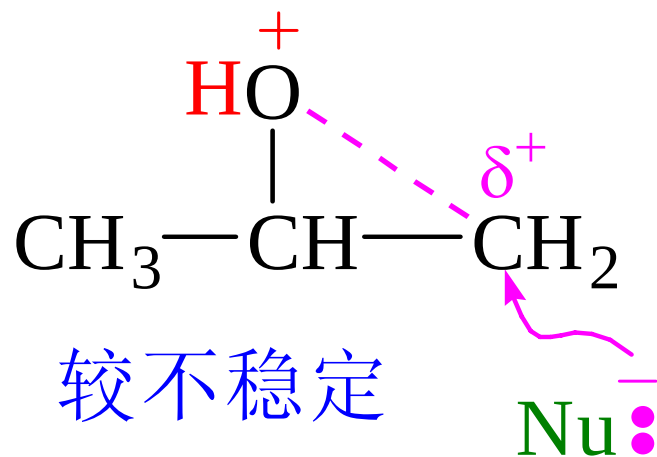
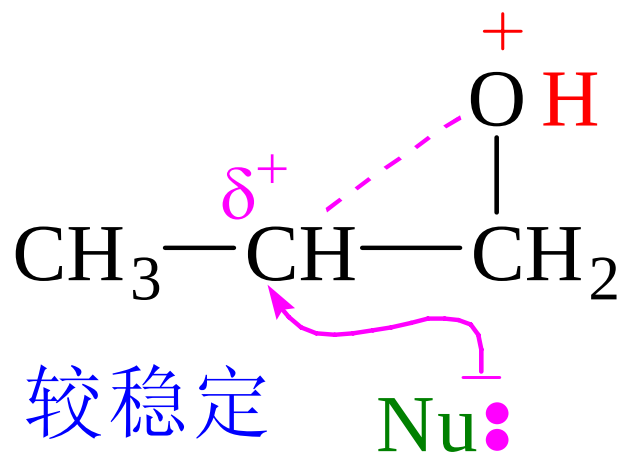
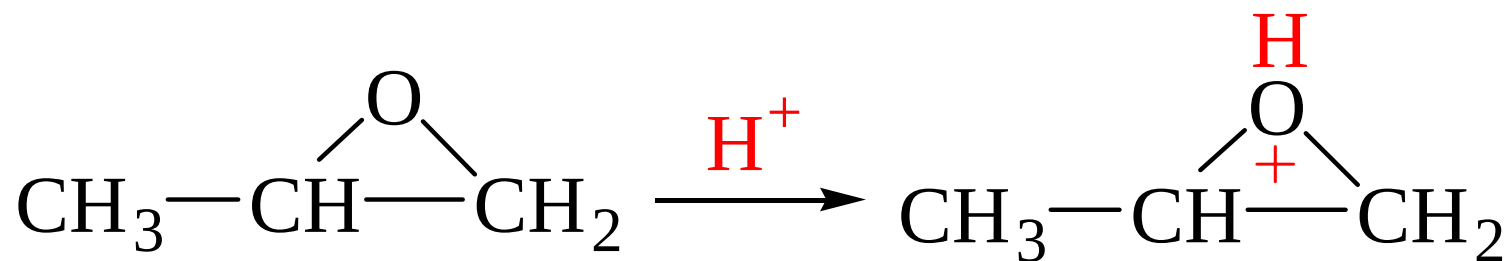
氢过氧化乙醚

- 检验：淀粉—碘化钾试纸
- 除去过氧化物的方法：FeSO₄洗涤
- 储存时加Fe可防止形成过氧化物

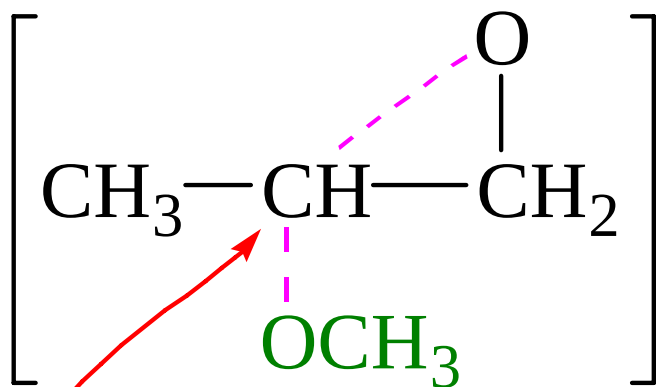
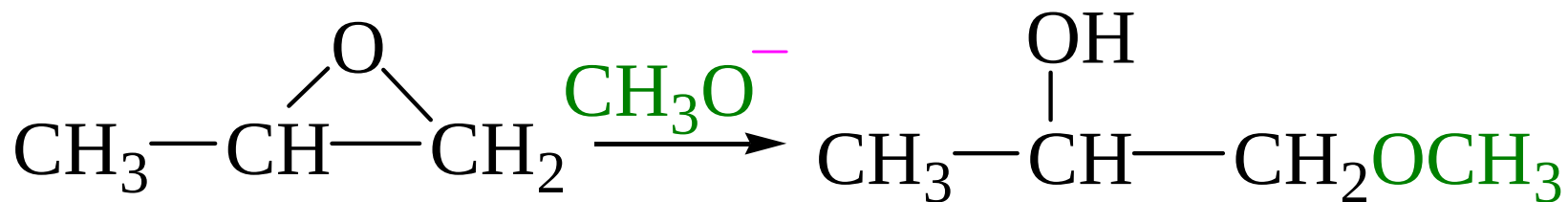
4. 环氧化合物的反应



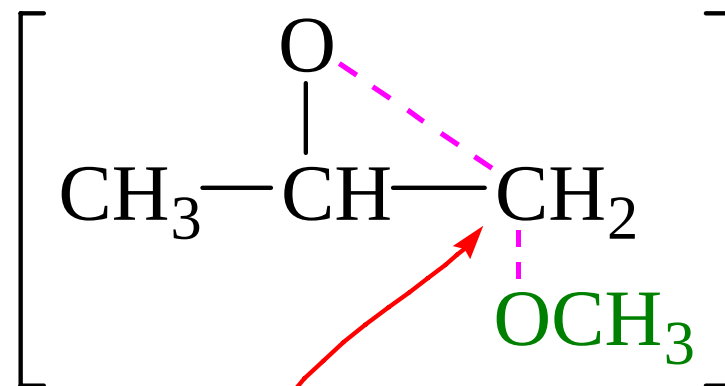
酸催化下的开环反应



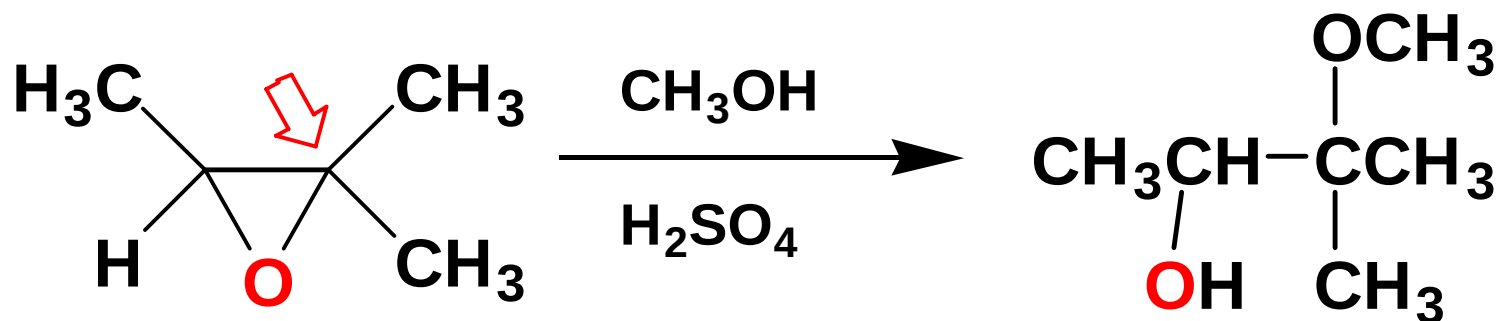
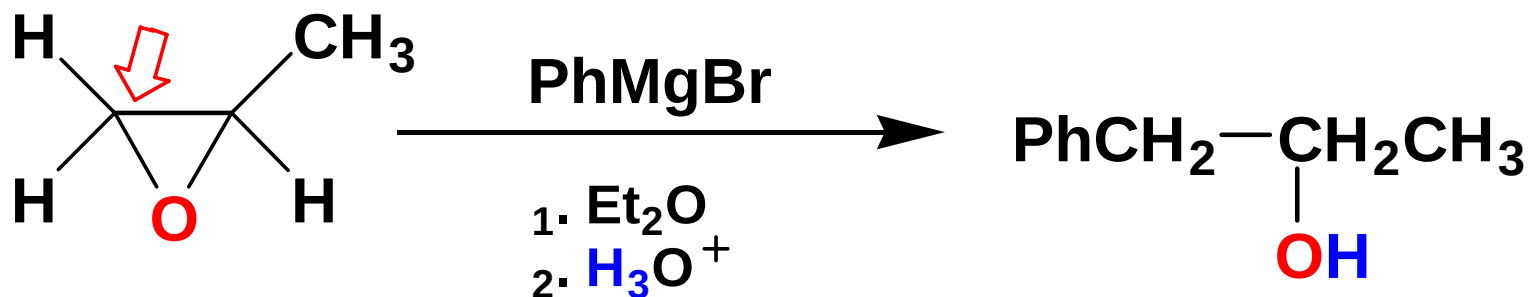
碱催化下的开环反应



位阻较大

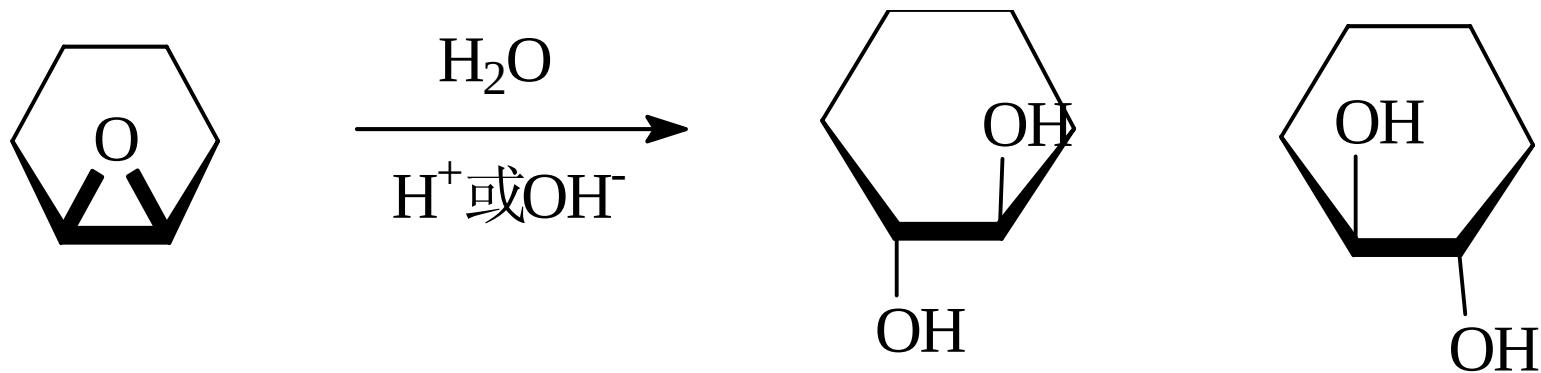
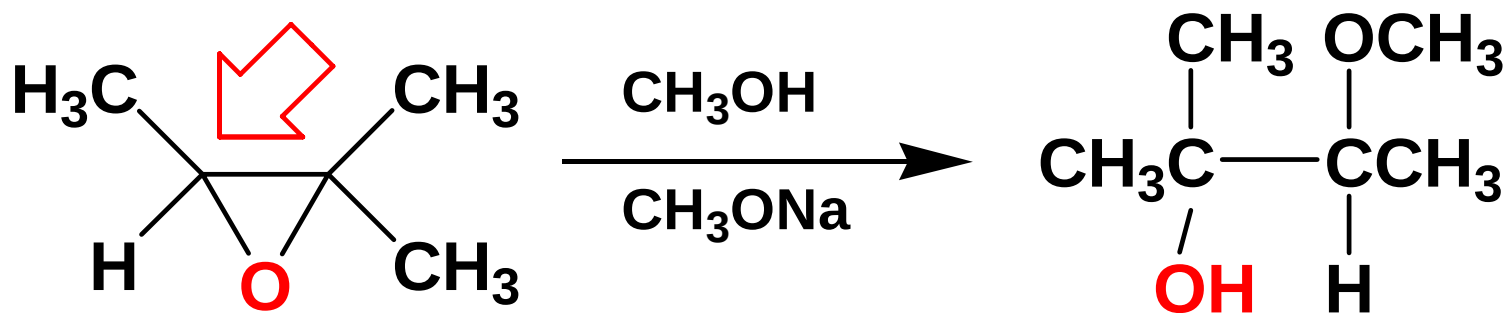


位阻较小



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Example



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作业

醇

- P165: 9.2
- P166: 9.3
- P167: 9.4
- P168: 9.5
- P169: 9.6
- P171: 9.7
- P173: 9.8

酚、醚

- P180: 9.10
- P184: 9.12
- P186: 9.13
- P189: 9.14

习题

P196: 3, 4,
5 (8)(9(10),
6, 7, 8,
9(1)